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# MTH 303 presentation 1 corrected

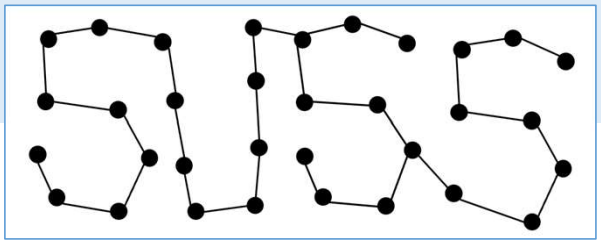
Principles of Graph Theory (Singapore University of Social Sciences)

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# MTH 303
# Principles of Graph Theory

## Study unit 1: Graphs and Digraphs





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# Two bees suffice



**Bee sceptical**

**Bee curious**





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# Assignments CMA01 and TMA01

- **CMA01**: online quiz covering study units 1 to 4. Opens on 15 Feb and closes on 22 Feb
- **TMA01**: tutor-marked assignment covering study units 1 to 5. It is available now on Canvas under the 'Assessments' tab. The date due is 28 Feb 2023. Please start working on the questions as we go through the topics each week. Do not wait till the moment as you need time to think through some of the questions.
- Please schedule your work carefully.
- Happy exploration and good luck.



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# GRAPH THEORY

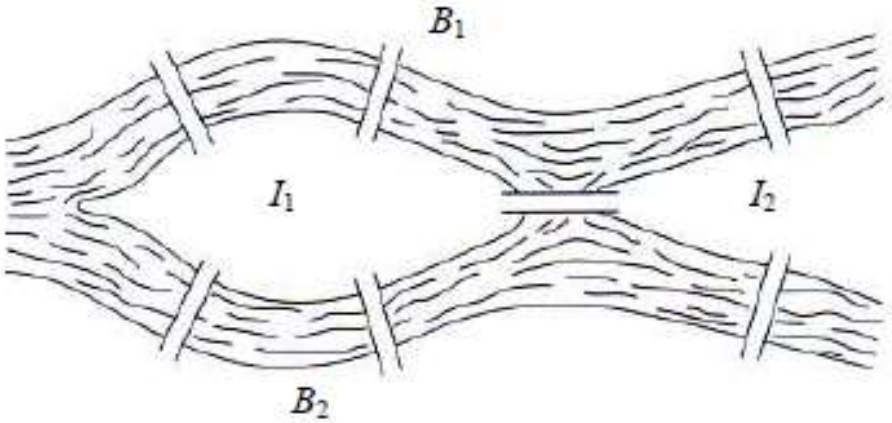
- Models relationships in many network optimization problems in industry and computer science, etc.
- Among topics in mathematics: more concrete, can draw pictures to help our intuition
- Helps us to improve our reasoning ability.
- Recreational aspects – can treat it as a game. To be a good player, we need to PRACTISE.
- Don't be afraid to dirty our hands, draw a lot of pictures and examples to get a feeling about the problems.



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# A) The seven bridges of Königsberg

- Is it possible to walk across each of the seven bridges once and only once?

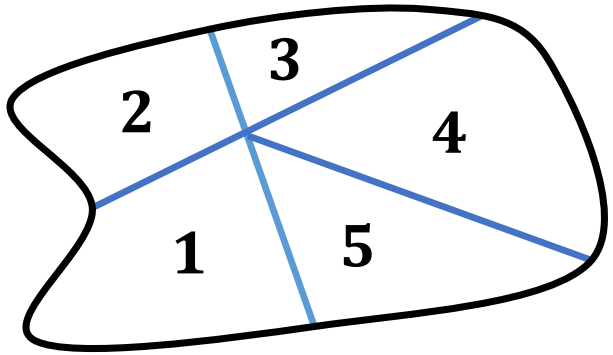




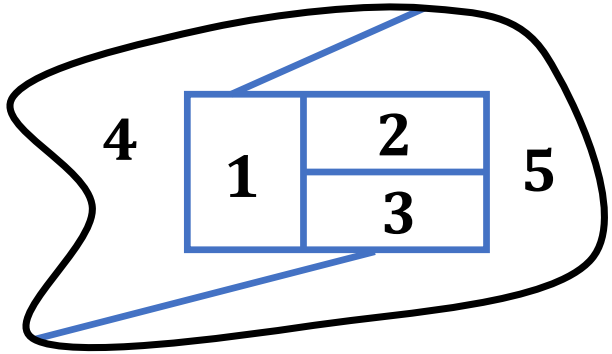

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# B) The king and his five sons

- A long time ago, there was a kingdom ruled by a king who had 5 sons. His wish was to divide his kingdom into 5 connected regions to be given to his 5 sons. The requirement was that every two region must share a common boundary. Can this be done?




Division 1: regions 3 and 5 share a common point but not common boundary



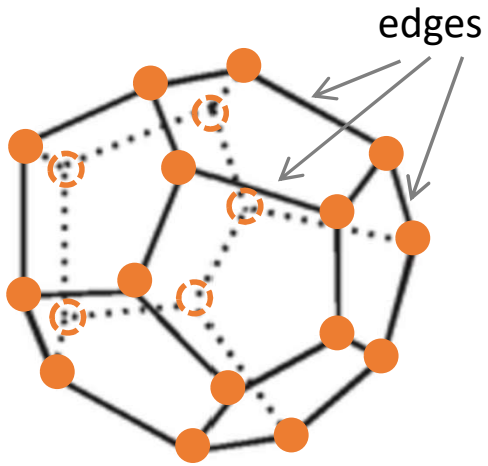

Division 2: does this work?



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# C) Round the world: dodecahedron

- The dodecahedron, shown below, has 20 corners highlighted by an orange dot. Each of these corners represent a city around the world. We can move from one city to a neighbouring city along an edge.
- Is it possible to make a tour of all the 20 cities and return to the starting city?






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# Outline

- Section 1: Graphs
  - Graphs and subgraphs
  - Degree of a vertex and degree sequences
  - Isomorphic graphs
  - Paths and cycles, connected graphs
  - Regular and bipartite graphs
  - Complement of a simple graph
- Section 2: Traversability
  - Eulerian and semi-Eulerian graphs
  - Hamiltonian graphs
  - Shortest paths in weighted graphs
- Section 3: Digraphs or directed graphs
  - Vertex degrees in digraphs
  - Isomorphic digraphs
  - Paths and cycles in digraphs
  - Eulerian and Hamiltonian digraphs
- Section 4: Matrix representations
  - Adjacency matrix
  - Incidence matrix



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# Section 1: Graphs

- 1) Graphs and subgraphs
  - adjacent vertices
  - subgraph
- 2) Degree of a vertex and degree sequences
- 3) Isomorphic graphs
- 4) Paths and cycles – connected graphs
- 5) Common families of graphs
  - regular graphs
  - bipartite graphs
- 6) Complement of a simple graph

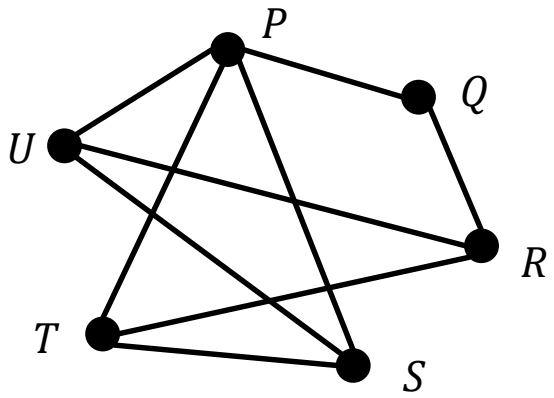


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# 1) Graphs and subgraphs

- A graph $G(V,\ E)$ has a vertex set $V$ and an edge set $E$.




- The dots $P, Q, R, S, T$ and $U$ are the **vertices** or **nodes.**
- Vertex set, $V = \{P, Q, R, S, T, U\}$

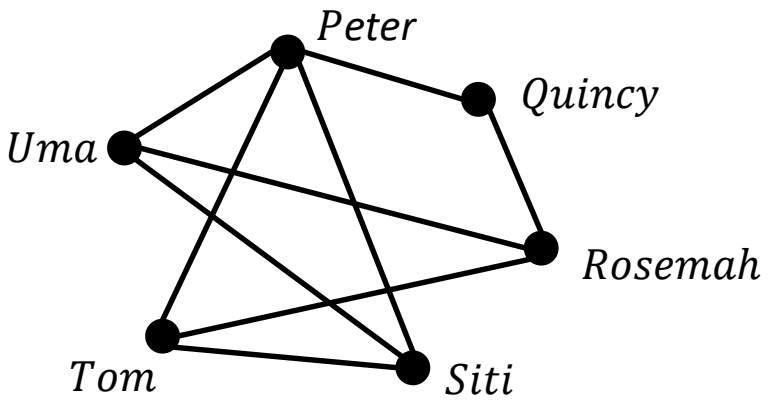
- An **edge** or **arc** joins two vertices. If the edge joins two vertices $P$ and $S$, we write the edge as $PS$ or $\{P, S\}$. Note that $\{P, S\} = \{S, P\}$ and the edge is **undirected**.
- There are 9 edges in the graph on the left.
- $E = \left\{ \begin{matrix} \{P.Q\}, \{P, S\}, \{P, T\}, \{P, U\}, \{Q, R\}, \\ \{R, T\}, \{R, U\}, \{S, T\}, \{S, U\} \end{matrix} \right\}$
- ~~$= \left\{ \begin{matrix} (P, Q),\ (P, S),\ (P, T),\ (P, U),\ (Q, R), \\ (R, T),\ (R, U),\ (S, T),\ (S, U) \end{matrix} \right\}$~~
- Note that the edge set $E \subseteq V \times V$.
- Sometimes we write the vertex and edge sets of $G(V, E)$ as $V(G)$ and $E(G)$ for emphasis.



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# Modelling friendships

- Vertices can represent people and the edge joins two people if they are friends.
- Eg. Uma and Siti are friends.
- Eg. Peter is not a friend of Rosemah.
- Eg. Tom and Rosemah are friends, etc.

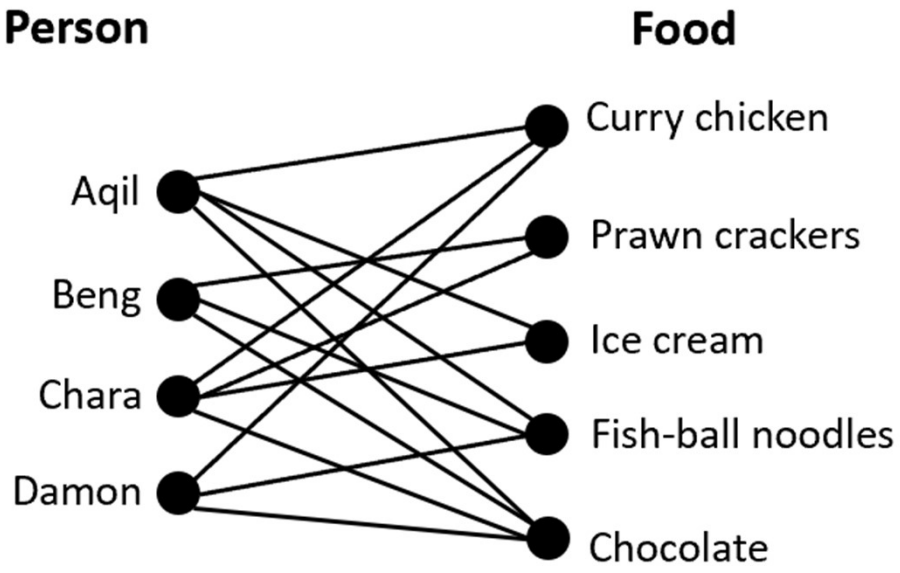





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# Another example

- In the previous example, the vertices represent people.
- In a graph, the vertices can represent different things.
- In the graph on the right, the vertices on the left represent people and those on the right represent types of food.
- There is an edge between a person and a food to indicate that the person likes the food.
- For example, Aqil and Chara like ice-cream.
- All four of them likes chocolate, etc.



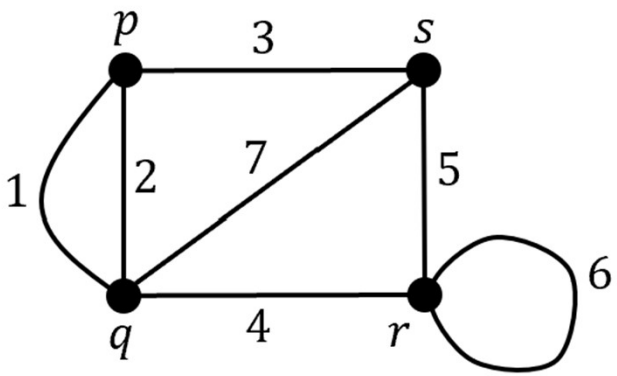



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# Order and size of a graph, multiple edges and loop

- **Order** of a graph = number of vertices
- **Size** of a graph = number of edges

- Order of $G$ = ?
- Size of $G$ = ?



Graph $G$

- **Multiple edges**: two vertices are connected by more than one edge. For example, edges 1 and 2 are multiple edges connecting vertices $p$ and $q$.
- **Loop**: an edge that connects a vertex to itself.
- Edge 6 is a loop. We write $6 = \{r, r\} = rr$.
- A **simple graph** is a graph without multiple edges and loops.



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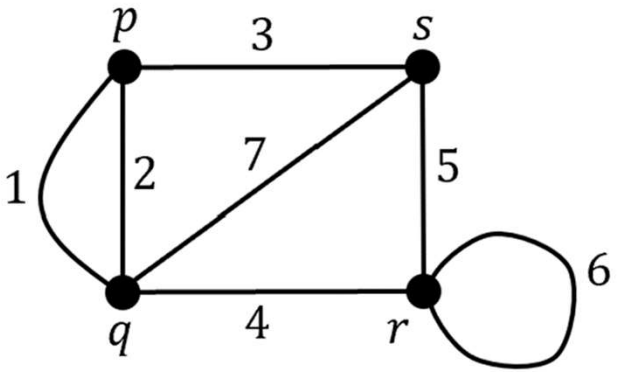
# Adjacency and incidence

- If two vertices $u$ and $v$ are joined by an edge $e$, we say that $u$ and $v$ are **adjacent** to each other. The edge $e$ is said to be **incident** to $u$ and $v$. Similarly, we say that vertices $u$ and $v$ are **incident** to edge $e$.

- Which vertices are adjacent to a) vertex $p$, b) vertex $r$?

- Which edges are incident to a) vertex $q$, b) vertex $r$, c) vertex $s$?

- Which vertices are incident to a) edge $2$, b) edge $4$, c) edge $6$?



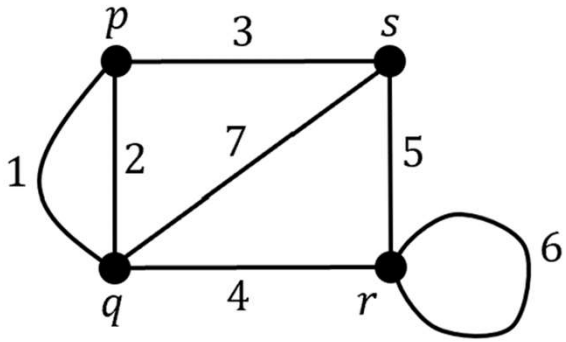
Graph $G$



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# Subgraph

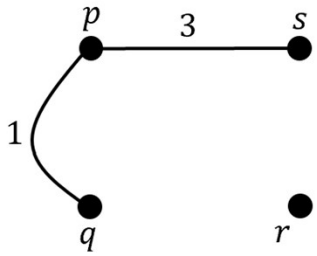
- Let $G$ be a graph with vertex set $V(G)$ and edge set $E(G)$.
- A graph $H$ is a **subgraph** of $G$ if $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$.
- The vertices and edges of $H$ are found in $G$.
- Consider the graph $G$ on the right.
- Below are examples of subgraphs of $G$




Graph $G$

$V(G) = \{p, q, r, s\}$

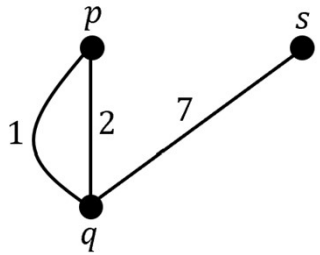
$E(G) = \{1, 2, 3, 4, 5, 6, 7\}$




Graph $H1$

$V(H1) = \{p, q, r, s\}$

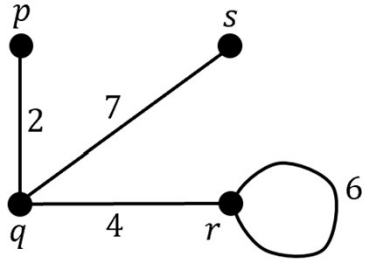
$E(H1) = \{1, 3\}$




Graph $H2$

$V(H2) = \{p, q, s\}$

$E(H2) = \{1, 2, 7\}$




Graph $H3$

$V(H3) = \{p, q, r, s\}$

$E(H3) = \{2, 4, 6, 7\}$

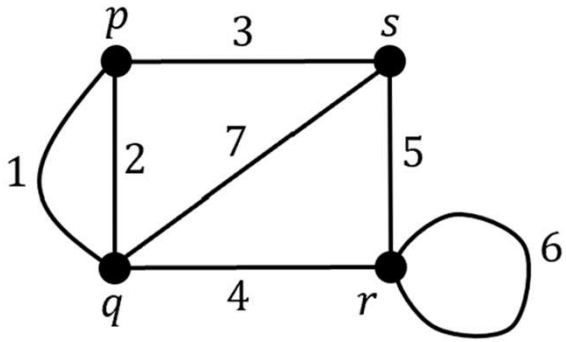


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# Spanning subgraph

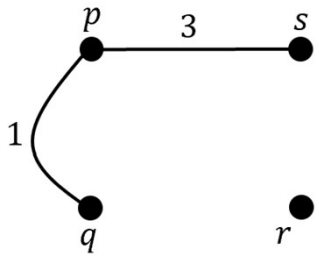
- Let $H(V(H), EH)$ be a subgraph of $G(V, E)$.
- $H$ is a **spanning subgraph** of $G$ if it has the same vertex set as $G$. That is, $V(H) = V$.
- Recall the examples of subgraphs from the previous slides
- Which of the subgraphs of $G$ is/are spanning subgraphs?




Graph $G$

$V(G) = \{p, q, r, s\}$

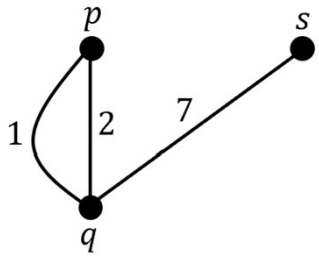
$E(G) = \{1, 2, 3, 4, 5, 6, 7\}$




Graph $H1$

$V(H1) = \{p, q, r, s\}$

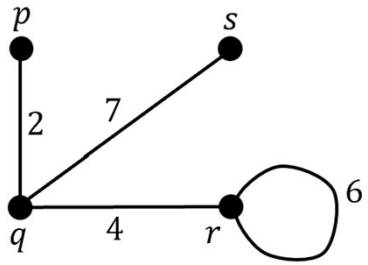
$E(H1) = \{1, 3\}$




Graph $H2$

$V(H2) = \{p, q, s\}$

$E(H2) = \{1, 2, 7\}$




Graph $H3$

$V(H3) = \{p, q, r, s\}$

$E(H3) = \{2, 4, 6, 7\}$



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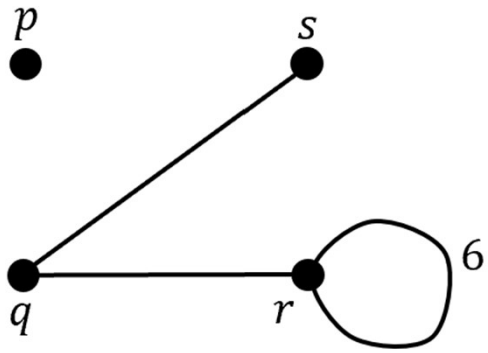
# 2) Degree of a vertex

- The number of edges incident to a vertex is an important number in many applications, and we give it a name. The **degree of vertex** $v$, denoted as $\boldsymbol{\deg(v)}$ is the number of edges incident to $v$. If there is a loop incident to $v$, then the loop contributes 2 to the degree of $v$.

- If $\deg(v)$ is an odd number, $v$ is called an **odd vertex** and it is an **even vertex** if $\deg(v)$ is even.

- If $\deg(v) = 0$, $v$ is called an **isolated vertex**.
- A vertex of degree 1 is called an **end–vertex** or a **leaf**.

- Compute the degrees of the vertices in $H4$
- Which are the odd vertices? Even vertices?
- Are there isolated vertices?
- Are there leaves in the graph?



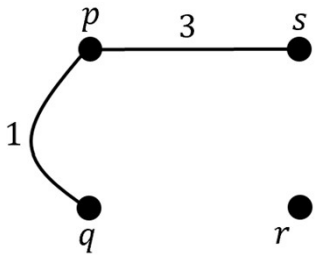
Graph $H4$



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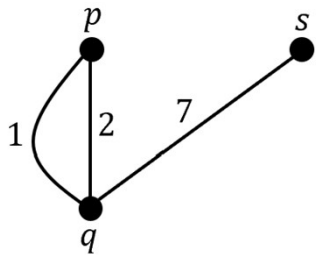
# Maximum and minimum degree

- The **minimum degree** of a graph $G$ is the smallest degree of its vertices. The minimum degree is denoted as $\boldsymbol{\delta(G)}$.
- The **maximum degree** of a graph $G$ is the largest degree of its vertices. The maximum degree is denoted as $\Delta(G)$.
- If $G$ is a simple graph of order $n$, then any vertex can be adjacent to at most $n-1$ vertices as there are no loops and multiple edges. Thus, $0 \leq \delta(G) \leq \Delta(G) \leq n-1$.
- A list of the degrees of the vertices in a graph is known as its **degree sequence**.



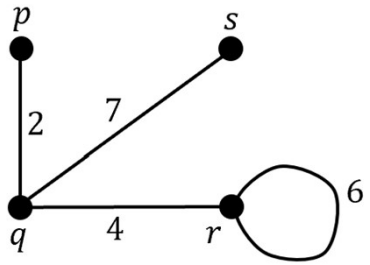

Graph $H1$

Degree sequence $= \{2,1,0,1\}$




Graph $H2$

Degree sequence $= \{2,3,1\}$




Graph $H3$

Degree sequence $= \{1,3,3,1\}$

What do we notice about the degree sequences?



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# Handshake lemma

- **Theorem 1.1 Handshake lemma**
- Let $G$ be a graph of order $n$ and size $m$. Let the vertices of $G$ be denoted by $v_1, v_2, \cdots, v_n$. Then the sum of the degrees of all the vertices equals to twice the number of edges. That is,

$$\sum_{i=1}^{n} \deg(v_i) = 2m.$$

- **Proof**
- If an edge $e$ joins two distinct vertices $v_i$ and $v_j$, then it contributes 1 to each of the $\deg(v_i)$ and $\deg(v_j)$. If $e$ is a loop incident to vertex $v_k$, then it contributes 2 to the degree of $v_k$, $\deg(v_k)$. In all cases, each edge contributes a total of 2 to the degrees of the vertices that the edge is incident to. Hence the result. ■
- **Corollary 1.1**
- In any graph, the number of vertices of odd degree is even.
- **Proof**: The sum of the degrees is an even number. Since the sum of an odd number of odd integers is odd, it follows that the number of odd degrees must be even.

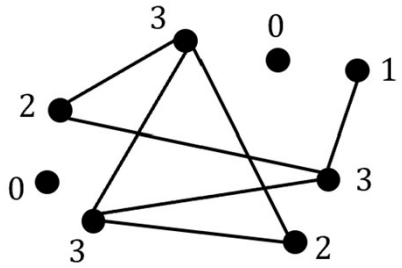


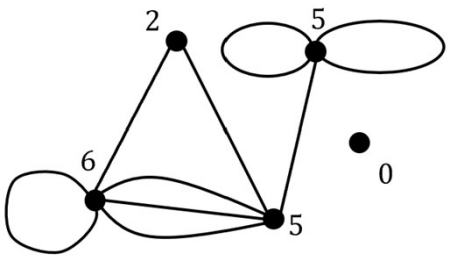
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# Degree sequences

- The **degree sequence** of a graph is obtained by listing the degrees of the vertices of the graph.
- The list may be arranged in a non-decreasing or non-increasing order or in any order.




Degree sequence $\{3, 2, 0, 3, 2, 3, 1, 0\}$ $\{2, 6, 5, 0, 5\}$

- Suppose we start with a sequence of non-negative integers, can we always find a graph whose degree sequence is the same as the given sequence?
- One necessary condition: the sum of the numbers must be even. Thus {2, 3, 4, 5, 6, 7} cannot be the degree sequence of a graph. Why?

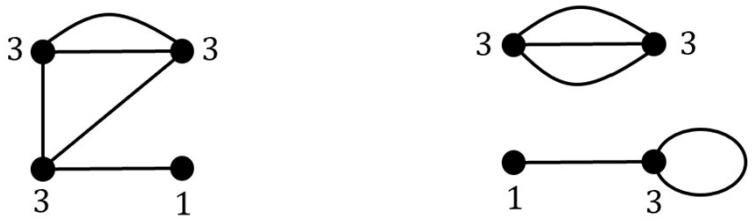


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# Graphical sequences

- A finite sequence of non-negative integers is called **graphical** if it is the degree sequence of a simple graph.

- Is {3, 3, 3, 1} graphical?




$$\text{degree sequence} = \{3, 3, 3, 1\}$$

- However we cannot find a simple graph whose degree sequence is $\{3, 3, 3, 1\}$. Convince yourself. So the sequence $\{3, 3, 3, 1\}$ is not graphical.
- Important to know the definitions well



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# Graphical sequences: a recipe

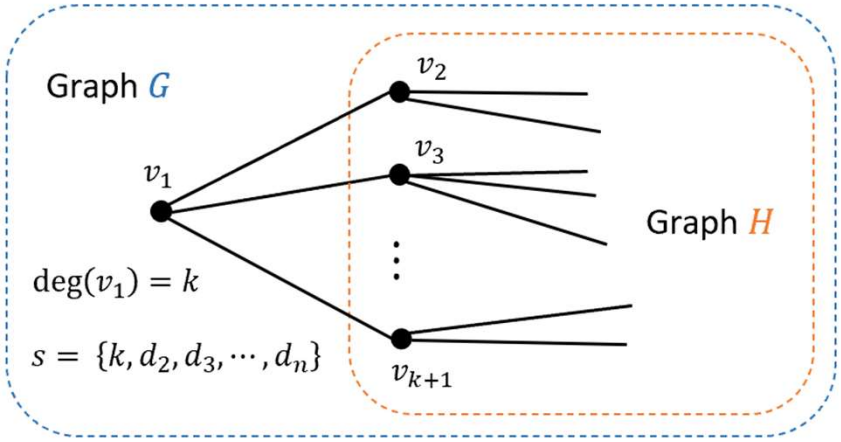
- Not easy to decide if a given sequence is graphical and the following theorem helps.
- **Theorem 1.2: Havel–Hakimi Theorem**
- A non-increasing sequence $s = \{k, d_2, d_3, \cdots, d_n\}$, $(n \geq 2)$, of non-negative integers, where $n > k \geq 1$, is graphical if and only if the sequence $\{d_2 - 1, d_3 - 1, \cdots, d_{k+1} - 1, \ d_{k+2}, \cdots, d_n\}$ is graphical.
- One direction of the result is easy to prove.
- Suppose the sequence $t = \{d_2 - 1, d_3 - 1, \cdots, d_{k+1} - 1, \ d_{k+2}, \cdots, d_n\}$ is graphical. Let $H$ be a simple graph with the degree sequence $t$ where vertices $v_2, v_3, \cdots, v_n$ have degrees $d_2 - 1, d_3 - 1, \cdots, d_n$ respectively.



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# Recipe in a picture

- Construct a simple graph $H$ having the degree sequence $t = \{d_2 - 1, d_3 - 1, \cdots, d_{k+1} - 1, d_{k+2}, \cdots, d_n\}$.
- Add a new vertex $v_1$ and add the edges $v_1v_2, v_1v_3, \cdots, v_1v_{k+1}$
- The resulting graph $G$ has degree sequence $s = \{k, d_2, d_3, \cdots, d_n\}$




$t = \{d_2 - 1, d_3 - 1, \cdots, d_{k+1} - 1, d_{k+2}, \cdots, d_n\}$



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# Aside: what “if and only if” means

- Let A and B be statements. In mathematics the statement A if and only if B means that A and B are equivalent. That is, A implies B and is in turn implied by B.
- Another way of saying this is, if A is true then B holds and if B is true that A is also true.
- To establish the result or theorem “A if and only if B”, we have to establish two things:
- i) we assume that A is true and show that B must also be true.
- ii) we show that if B is true then it follows that A must also be true.

- In the example,
- A is the statement: $s = \{k, d_2, d_3, \cdots, d_n\}$ is graphical, and
- B is the statement: $t = \{d_2 - 1, d_3 - 1, \cdots, d_{k+1} - 1,\ d_{k+2}, \cdots, d_n\}$ is graphical.
- Our diagram sketches out the argument B implies A. That is, if B is true that A is also true.
- We will not show i) A implies B.



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# Applying the theorem

- Let's try to figure out whether the sequence $\{3, 2, 1, 5, 4, 1, 4\}$ is graphical.
- First arrange it in non-increasing order $S = \{5, 4, 4, 3, 2, 1, 1\}$.
- Using the recipe, we remove the integer 5 and then delete 1 from the subsequent five numbers to obtain $T = \{4 - 1, 4 - 1, 3 - 1, 2 - 1, 1 - 1, 1\} = \{3, 3, 2, 1, 0, 1\} = \{3, 3, 2, 1, 1, 0\}$
- Is the sequence $T = \{3, 3, 2, 1, 1, 0\}$ graphical? If it is, then $S = \{5, 4, 4, 3, 2, 1, 1\}$ is also graphical.
- We apply the recipe again to obtain $U = \{3 - 1, 2 - 1, 1 - 1, 1, 0\} = \{2, 1, 1, 0, 0\}$.
- We can see that $U = \{2, 1, 1, 0, 0\}$ is graphical. If we cannot see that, just apply the recipe again.
- We apply the theorem again and obtain the sequence $V = \{1 - 1, 1 - 1, 0, 0\} = \{0, 0, 0, 0\}$ which is clearly graphical.

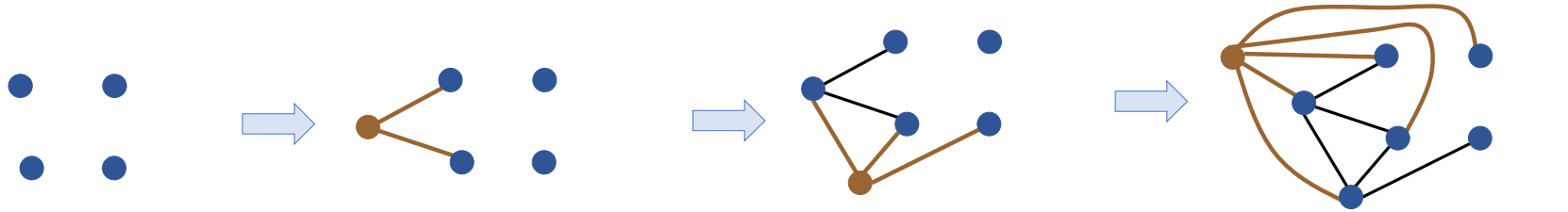


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# Constructing the simple graph iteratively

- $S = \{5, 4, 4, 3, 2, 1, 1\} \rightarrow T = \{3, 3, 2, 1, 1, 0\} \rightarrow U = \{2, 1, 1, 0, 0\} \rightarrow V = \{0, 0, 0, 0\}$
- What graph has $V = \{0, 0, 0, 0\}$ as its degree sequence? A graph with four isolated vertices of course. So that's where we start.



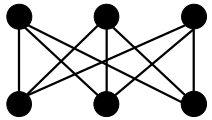
$V = \{0, 0, 0, 0\}$ $U = \{2, 1, 1, 0, 0\}$ $T = \{3, 3, 2, 1, 1, 0\}$ $S = \{5, 4, 4, 3, 2, 1, 1\}$



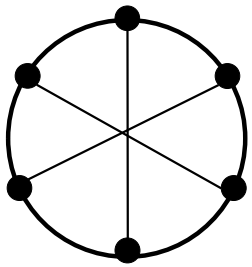
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# 3) Isomorphic graphs

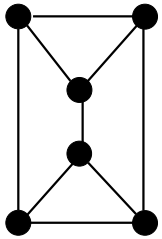
- What is important in a graph is the connection between vertices. How it is drawn on paper should not matter. The notion of isomorphism captures this idea. The word comes from the Greek language; ‘isos’ meaning same and ‘morphe’ meaning shape or structure.
- Consider the four graphs below. Which of these graphs are equivalent or essentially the same? That is, which of them are a re-drawing of another? Take some time to figure it out before looking at the answer.



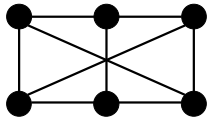
Graph $G1$



Graph $G2$



Graph $G3$



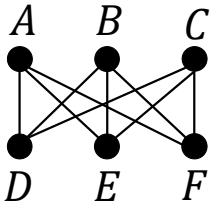
Graph $G4$



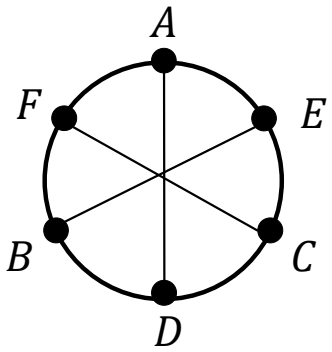
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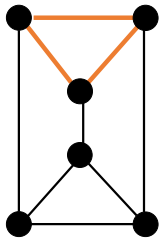
# Here's the answer



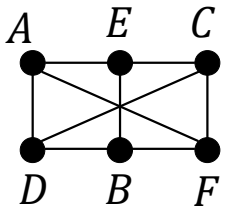

Graph $G1$




Graph $G2$



Graph $G3$




Graph $G4$

- With the labelling, it is clear that the three graphs $G1$, $G2$ and $G4$ represent the same graph. So they are isomorphic to each other.
- Graph $G3$ is different from the other three. It has a triangle as shown by the orange edges and none of $G1$, $G2$ and $G4$ has three edges that form a triangle. So $G3$ is not isomorphic to any of the other graphs.
- In fact, $G1$, $G2$ and $G4$ have only even cycles. They are bipartite graphs (see later section).



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# Isomorphic graphs

- Let $G(V_G, E_G)$ and $H(V_H, E_H)$ be two graphs. We say that $G$ and $H$ are **isomorphic** if there is a bijection $\varphi$ (one–to–one and onto function) that maps $V_G$ onto $V_H$ such that for any two vertices $p, q \in V_G$, we have $pq \in E_G$ if and only if $\varphi(p)\varphi(q) \in E_H$.
- The function $\varphi$ is known as an isomorphism from $G$ to $H$.

- If graphs $G$ and $H$ are isomorphic, then they have
- i) the same order (number of vertices),
- ii) the same size (number of edges),
- iii) the same degree sequence.
- If any of i) or ii) or iii) is not satisfied, the two graphs cannot be isomorphic.
- Isomorphic graphs have the same number of triangles, etc.

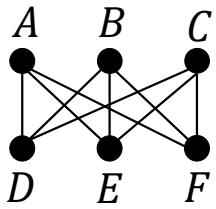


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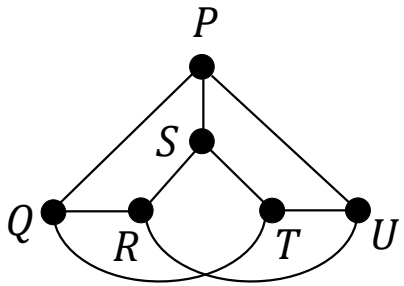
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# An obvious result

- **Theorem 1.3**
- Let $\varphi$ be an isomorphism from graph $G$ to $H$. For any vertex $v \in V_G$, $\deg(v) = \deg\big(\varphi(v)\big)$. That is, the degree of $v$ is equal to the degree of its image $\varphi(v)$under $\varphi$.
- Example: are graphs $G$ and $H$ isomorphic?




Graph $G$




Graph $H$

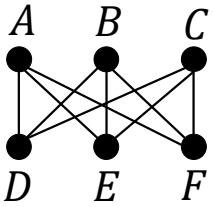
- How can we prove that they are isomorphic? Try to find an isomorphism to identify the vertices that preserve the edge structure.



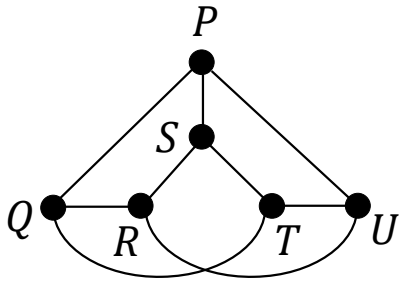
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# The mapping

- Example: are graphs $G$ and $H$ isomorphic?




Graph $G$




Graph $H$

- Consider the function $\varphi$: $V_G \rightarrow V_H$

  where $\varphi(A) = P, \varphi(B) = R, \varphi(C) = T, \varphi(D) = Q, \varphi(E) = S, \varphi(F) = U.$

- It is easy to check that $\varphi$ is an isomorphism. To check that for every pair of vertices $l$ and $m$ in graph $G$, $lm$ is an edge in $G$ if and only if the image $\varphi(l)\varphi(m)$ is an edge in $H$.
- For example, $AD$ is an edge in $G$ and $\varphi(A)\varphi(D) = PQ$ is also an edge in $H$.
- For example, $AC$ is not an edge in $G$ and $\varphi(A)\varphi(C) = PT$ is also not an edge in $H$, and so forth.
- Thus $\varphi$ is a bijection that preserves the edge structure. So it is an isomorphism.



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# 4) Paths and cycles

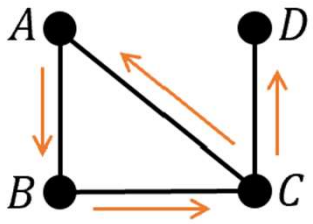
- **Walk**: a **walk of length $k$** in a graph is a succession of edges of the form $v_1v_2, v_2v_3, \cdots, v_kv_{k+1}$. Note that $v_iv_{i+1}$ is an edge in the graph for $i = 1, 2, \cdots, k$. There are $k$ edges in a walk of length $k$.
- Note that the vertices $v_i$ may be repeated.
- As a shorthand notation, we will just write the above as $v_1v_2v_3 \cdots v_{k+1}$ and refer to it as a **walk between $v_1$ and $v_{k+1}$**. This walk is also called a **$v_1 - v_{k+1}$ walk**.

- **Trail**: a **trail** is a walk where all the edges in it are distinct. Note: the vertices may not be distinct.

- **Path**: a **path** a walk where all the vertices in it are distinct. In particular, the start and end vertices are distinct.

- Is a path always a trail? What about the converse?

- In the graph on the right, is the sequence $CABCD$ a trail?
- Is it a path?

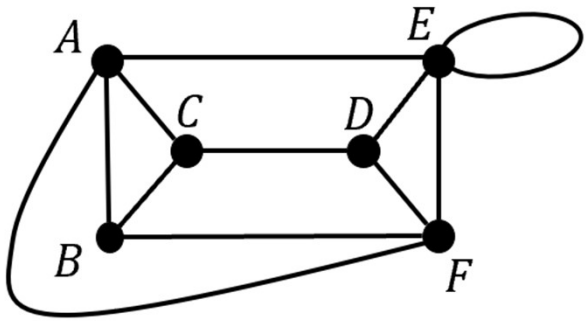


$CABCD$



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# Example



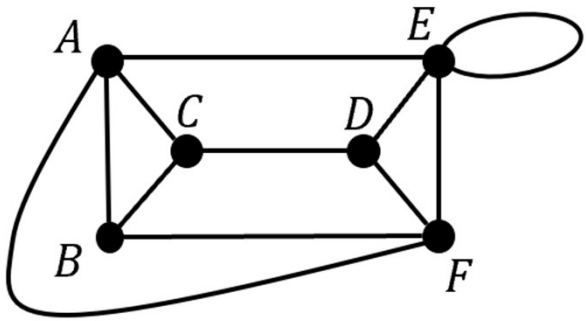

- In the graph above, please identify whether the following sequences constitute a walk, trail, path by placing a tick ✓ in the appropriate box.

| | Walk | Length of walk | Trail | Path |
|---|---|---|---|---|
| CAEFDEABFEEA | | | | |
| BFEDFACD | | | | |
| CDEEABDCA | | | | |
| DCBFEEDFAED | | | | |
| ACBFEFBAED | | | | |



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# Example: answer




- In the graph above, please identify whether the following sequences constitute a walk, trail, path by placing a tick ✓ in the appropriate box.

| | Walk | Length of walk | Trail | Path |
|---|---|---|---|---|
| CAEFDEABFEEA | ✓ | 11 | × | × |
| BFEDFACD | ✓ | 7 | ✓ | × |
| CDEEABDCA | × | | × | × |
| DCBFEEDFAED | ✓ | 10 | × | × |
| ACBFEFBAED | ✓ | 9 | × | × |



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# A $P - S$ walk always contains a $P - S$ path

- **Theorem 1.4**
- Let $P$ and $S$ be distinct vertices in a graph. Then every $P - S$ walk will contain a $P - S$ path.
- **Proof**
- Suppose $W = V_1V_2 \cdots V_kV_{k+1}$ is a $P - S$ walk of length $k$ in the graph. Note that $P = V_1$ and $S = V_{k+1}$. If all the vertices $V_i$ are distinct, then the walk $W$ is the required path.
- Otherwise, we have $V_i = V_j$ for some $1 \leq i < j \leq k + 1$.
- We delete or trim the vertices $V_{i+1} \cdots V_j$ in the sequence for $W$ to obtain a shorter $P - S$ walk $W' = V_1 \cdots V_iV_{j+1} \cdots V_{k+1}$.
- If all the vertices in $W'$ are distinct then it is a required path from $P$ to $S$.
- Otherwise, we can repeat the trimming process to obtain a shorter $P - S$ walk. Since the walk $W$ has finite length, we eventually end up with a $P - S$ path as required. ■
- Example: In the walk $ACBFEFBAED$ in a graph, find a $A - D$ path.



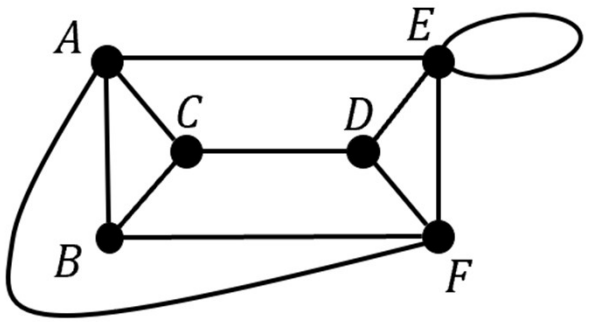
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# Closed walk, circuit and cycle

- A walk of length at least one is **closed** if the initial vertex and final vertex are the same.
- A **closed trail** is a closed walk in which all the edges are distinct. A closed trail is also called a **circuit**.
- A **cycle** is a closed walk where the vertices are distinct except for the initial and final vertex which are the same.
- Classify the sequences in the table below with reference to the graph on the right of the table.

| | Closed Walk | Circuit | Cycle |
|---|---|---|---|
| CAEFDEAC | | | |
| BFEDFACB | | | |
| EFDCBAE | | | |
| FBCAEEDF | | | |



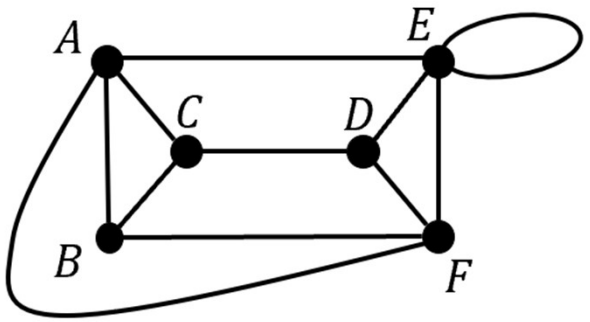



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# Closed walk, circuit and cycle

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- A **cycle** is a closed walk where the vertices are distinct except for the initial and final vertex which are the same.
- Classify the sequences in the table below with reference to the graph on the right of the table.

| | Closed Walk | Circuit | Cycle |
|---|---|---|---|
| CAEFDEAC | ✓ | | |
| \\BFEDFACB | | | |
| EFDCBAE | | | |
| FBCAEEDF | | | |

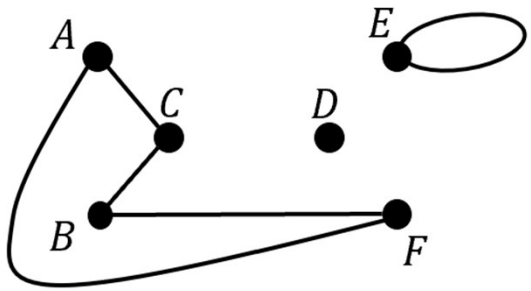





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# Connected graphs

- A graph is said to be **connected** if every two vertices are connected by a path. Otherwise it is said to be **disconnected**.
- Each graph is made up of one or more connected **components** (connected pieces).
- How many components does a connected graph have?
- Example: the following graph is not connected and has three (connected) components.

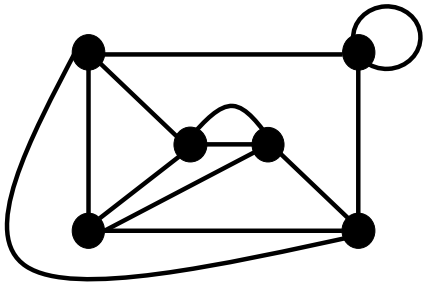




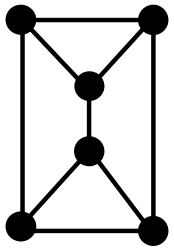

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# 5) Some families of graphs: regular graph

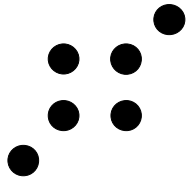
- **Regular graphs**: a graph is **regular** if all its vertices have the same degree. If all the vertices of the graph $G$ have degree $r \geq 0$, then $G$ is ***r*-regular** or regular of degree $r$.



4-regular graph



3-regular graph



0-regular graph

- **Theorem 1.5**
- A $r$-regular graph on $n$ vertices has exactly $^{nr}/_{2}$ edges.
- **Corollary**
- There is no $r$-regular graph on $n$ vertices where both $r$ and $n$ are odd integers.

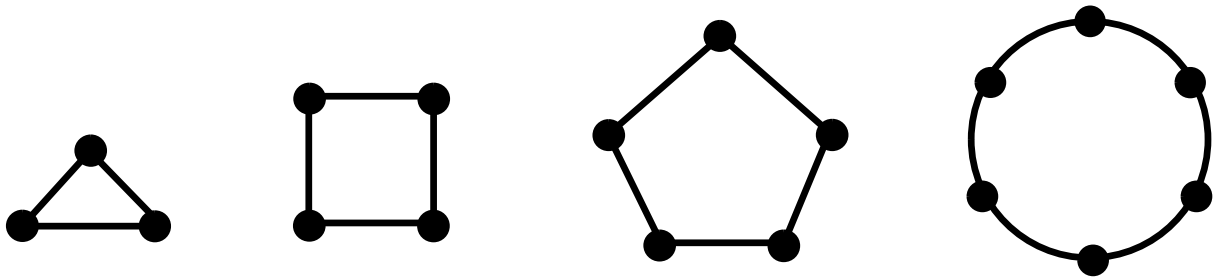


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# Cycles

- A **cycle** graph is a graph consisting of a single cycle of vertices linked by edges. The cycle graph with $n$ vertices is denoted by $C_n$.
- Obviously, a cycle graph is a 2-regular graph. Some cycle graphs are shown below.



$C_3$ $C_4$ $C_5$ $C_6$

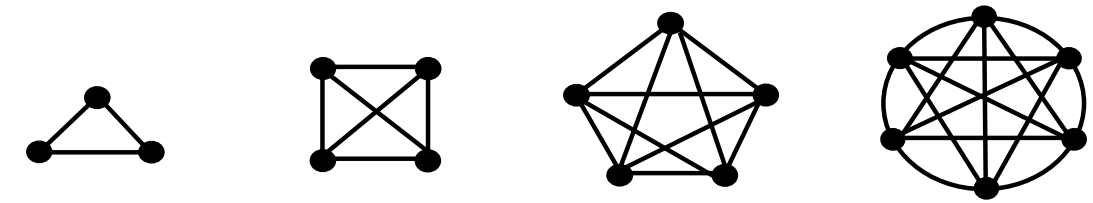


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# Complete graphs

- A **complete graph** is a simple graph in which every two distinct vertices are joined by exactly one edge.
- The complete graph with $n$ vertices is denoted as $K_n$.
- Therefore, $K_n$ has the maximum possible number of edges for a simple graph with $n$ vertices.
- Since every two distinct vertices of $K_n$ are joined by an edge, the number of edges in $K_n$ is

$$\binom{n}{2} = \frac{n(n-1)}{2}$$



$C_3 = K_3$ $K_4$ $K_5$ $K_6$

- Note that $K_n$ is a regular graph of degree $n - 1$.

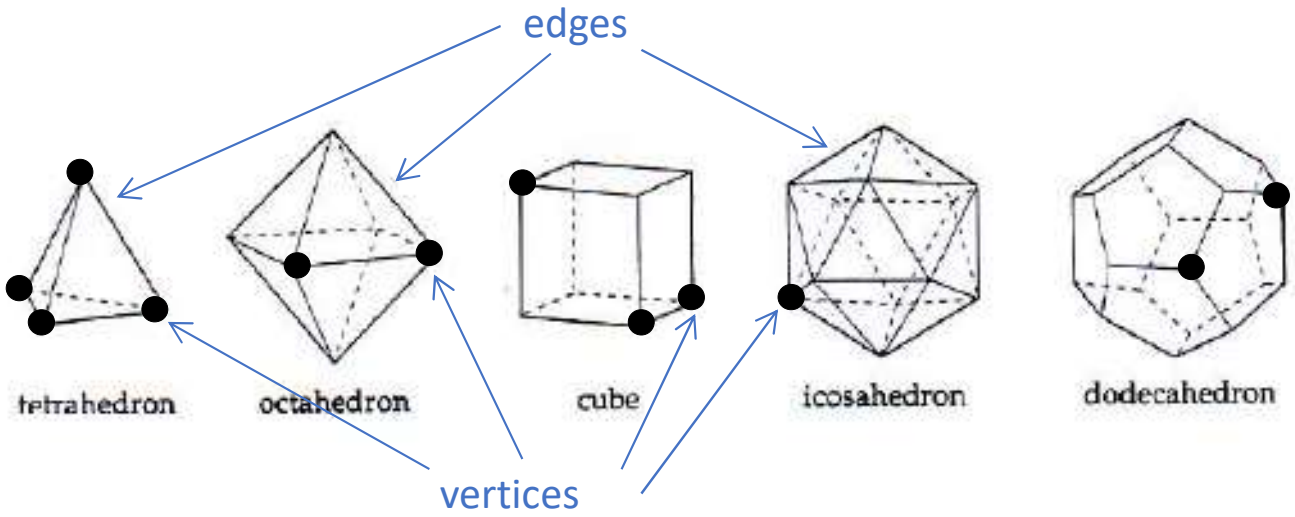


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# Platonic graphs

- Platonic graphs are related to the five platonic solids shown below.



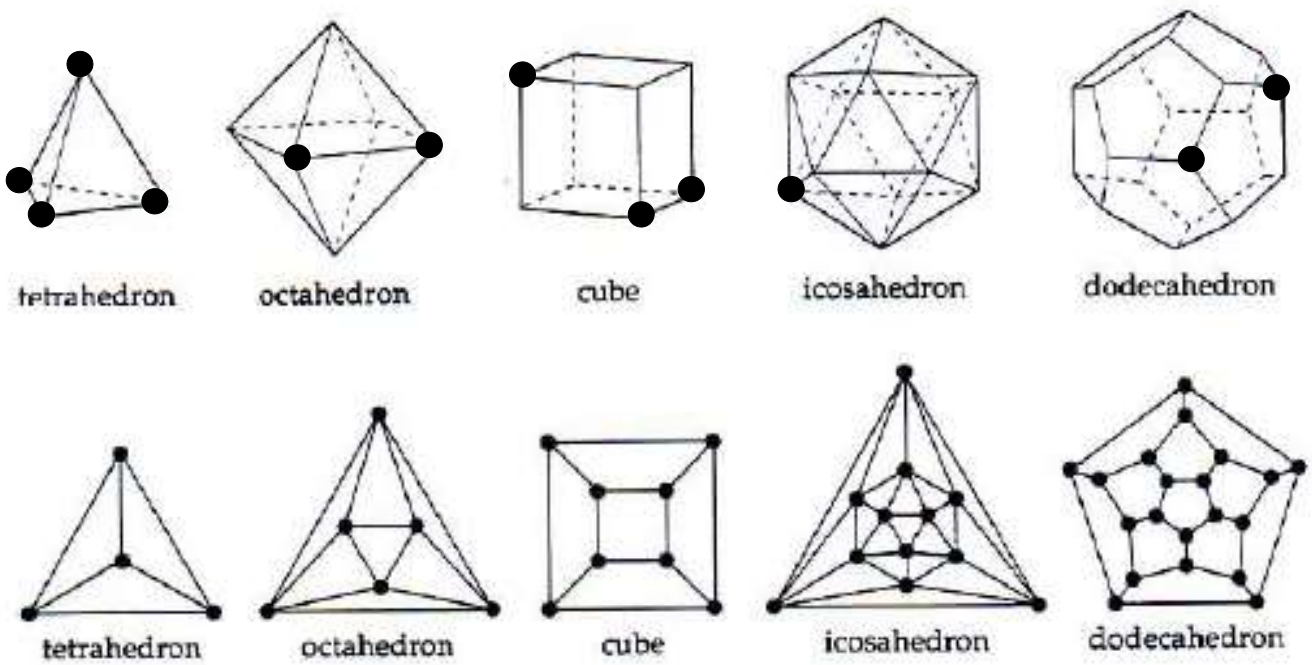

- The vertices in the graph are the corners of the solids, and the edges in the graph are the edges in the solid.



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# Platonic graphs can be drawn on a plane

- The five Platonic graphs are given below. Note that they are all regular graphs.

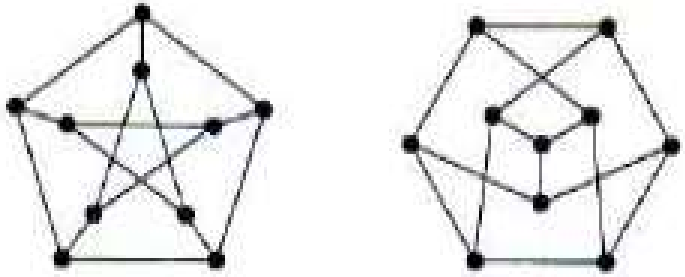





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# Petersen graph

- It is a 3–regular graph with 10 vertices and 15 edges as shown below.



Two common drawings of the Petersen graph

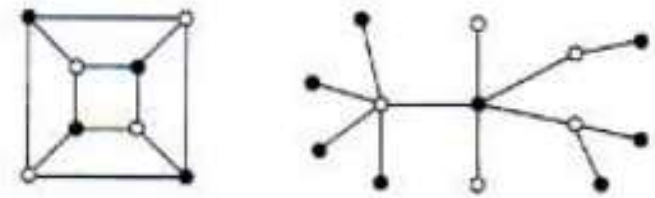
- The Petersen graph has interesting properties and is a counter-example to several conjectures.



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# Bipartite graphs

- If the vertices of a graph $G$ can be partitioned into two sets $X$ and $Y$ such that every edge of $G$ joins a vertex in $X$ to a vertex in $Y$, we call $G$ a bipartite graph.
- Below are some examples of bipartite graphs



- The black and white vertices form a partition of the vertex set. We see that every edge joins a white vertex to a black vertex.



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# Characterization of a bipartite graph

- **Theorem 1.6**
- A graph $G$ is bipartite if and only if it does not contain an odd cycle.

- How can we prove such a statement.
- **Proof**
- It is enough to prove that each component of $G$ is bipartite. Without loss of generality, we may assume that $G$ has one component.
- ($\Rightarrow$) Suppose $G$ is bipartite and $(X, Y)$ is a partition of the vertex set $V(G)$. Consider any cycle $C$: $x_1 y_1 x_2 y_2 \cdots x_k y_k x_1$ where the vertices $x_i \in X$ and $y_j \in Y$. Since the vertices have to alternate between $X$ and $Y$, we see that the cycle $C$ has $2k$ edges and it is an even cycle.
- ($\Leftarrow$) Let $v$ be any vertex in $G$.
- Define $X = \{x \in V(G)$: a path from v to x has even number of edges$\}$. Let $Y = V(G) \setminus X$.



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# Proof of Theorem 1.6

- Since $G$ has no odd cycles, the sets $X$ and $Y$ are well defined. Suppose on the contrary, there is a vertex $w \in X \cap Y$. Then there is a path $P: vu_1 \cdots u_t w$ of even length and a path $Q: vs_1 \cdots s_r w$ of odd length. Then we have a closed walk $vu_1 \cdots u_t w s_r \cdots s_1 v$ of odd length which must contain an odd cycle, which is a contradiction.
- Moreover every edge must join a vertex in $X$ to a vertex in $Y$. Thus $G$ is bipartite and $(X, Y)$ is associated partition of the vertex set.

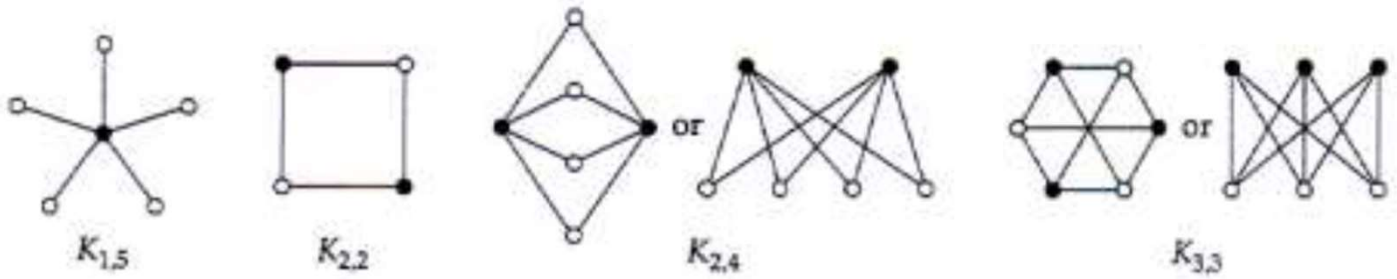


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# Complete bipartite graphs

- A **complete bipartite graph** is a bipartite graph with partition $X$ and $Y$ in which every vertex in $X$ is joined to every vertex in $Y$ by exactly one edge.
- A complete bipartite graph with $|X| = s$ and $|Y| = t$ is denoted by $K_{s,t}$ or $K_{t,s}$. The number of edges in $K_{s,t}$ or $K_{t,s}$ is equal to $st$.
- Some examples of complete bipartite graphs are:

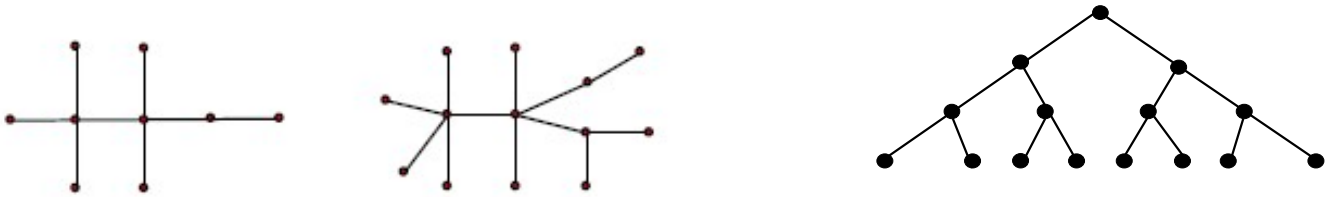





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# Trees

- A **tree** is a simple connected graph with no cycles. Trees are bipartite graphs.
- Some examples of trees



- In a tree, every two vertices are connected by a unique path. A tree on $n$ vertices has exactly $n-1$ edges.
- In a tree, a vertex of degree one is known as a **leaf.**
- A **path graph** is a tree with exactly two vertices of degree one. A path with $n$ vertices is denoted by $P_n$.

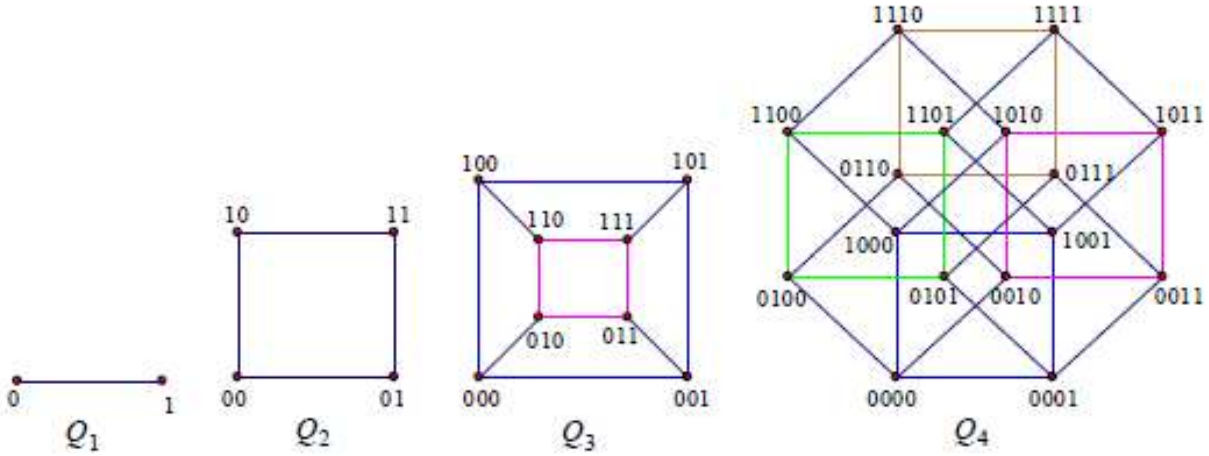





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# Cubes

- Example of cubes




- The **cube** $Q_n$ has $2^n$ vertices where each vertex is represented by string of $n$ symbols $a_1 a_2 \cdots a_n$ with $a_i \in \{0, 1\}$.

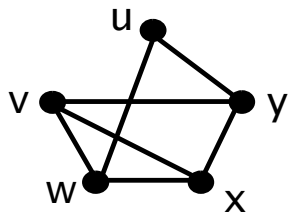
- Two vertices are connected by an edge if and only if they differ in exactly one co-ordinate.



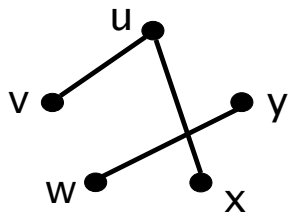
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# 6) Complement of a simple graph

- For a simple graph $G$, its complement $\bar{G}$ is a graph whose vertex set $V(\bar{G}) = V(G)$ is the same as that of $G$ and its edge set $E(\bar{G}) = \{uv | u, v \in V(\bar{G}), uv \notin E(G)\}$.
- The complement $\bar{G}$ has the same vertex set as $G$. An edge $uv$ is in $\bar{G}$ if and only if the edge $uv$ is not in $G$.




**Graph $G$**




**Graph $\bar{G}$**

- Notice that the edges of $G$ and its complement $\bar{G}$ combined will give the complete graph.
- What is the complement of the complement? That is, what is $\overline{(\bar{G})}$?



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# Complement of a simple graph, cont'd

- **Theorem 1.7**
- If $G$ is disconnected, then its complement $\bar{G}$ is connected.

- **Proof**:
- Consider any two vertices $u$ and $v$ in $\bar{G}$.

- Case 1:
- If $u$ and $v$ are in different components in $G$, then the edge $uv \in E(\bar{G})$.

- Case 2:
- Suppose $u$ and $v$ are in the same component in $G$. Since $G$ is not connected, it has a component that does not contain $u$ or $v$ and we pick a vertex $w$ from this component. Note that the edges $uw$ and $vw$ are not in $G$. Thus the edges $uw$ and $vw$ are in $E(\bar{G})$ .

- The vertices $u$ and $v$ are connected by a path in $\bar{G}$ in all cases.
- Thus the complement $\bar{G}$ is connected. ■



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# Complement of a simple graph, cont'd

- Another result that is easy to see is the following

- **Theorem 1.8**
- Two simple graphs $G$ and $H$ are isomorphic if and only if their complements are isomorphic.

- In practice, it may be difficult to show that two graphs $G$ and $H$ are isomorphic if both contain many edges or are drawn in a confusing manner. Using this theorem, we can change the problem to checking that their complements are isomorphic, which hopefully is less messy.



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# Section 2: Traversability

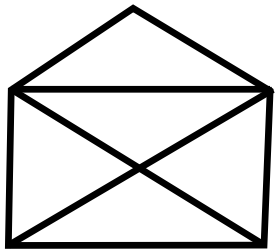
- 1) Eulerian graphs and semi-Eulerian graphs
- 2) Hamiltonian graphs
- 3) Shortest paths
  - Shortest paths in weighted graphs



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# 1) Eulerian and semi-Eulerian graphs

- Trace the figure below without lifting your pen from the paper, starting from any point subject to the condition that each segment is traced only once.



- Can it be accomplished? What is the key observation?

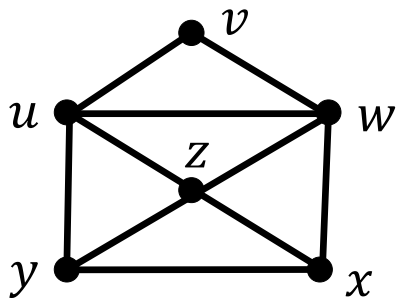


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# Eulerian graphs

- For a given graph $G$, a trail in $G$ that contains all the edges is called an **Eulerian trail**.
- A circuit of $G$ that contains all the edges is called an **Eulerian circuit** of $G$.
- A graph $G$ is called an **Eulerian graph** if it has an Eulerian circuit.
- If $G$ has an open Eulerian trail, but not an Eulerian circuit, it is called a **semi–Eulerian graph**.
- Note: an Eulerian graph is not a semi-Eulerian graph by the above definition.




- Is the above graph Eulerian or semi-Eulerian or neither?



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# Characterisation of Eulerian graphs

- **Theorem 2.1**
- A connected graph $G$ is Eulerian if and only if each vertex in $G$ has even degree.

- Idea of proof: we can construct a Eulerian circuit starting at any vertex and we will not be stuck since every vertex has even degree. So we eventually end up at the starting vertex. If there are still edges that are not on the circuit, we can extend the circuit to include them.

- **Corollary 2.1**
- A connected graph $G$ is semi-Eulerian if and only if exactly two vertices in $G$ have odd degrees. Furthermore, each Eulerian trail of $G$ will originate from one of these odd vertices and end at the other.

- **Corollary 2.2**
- A connected graph $G$ is Eulerian if and only if $G$ is a disjoint union of edge disjoint cycles in $G$.

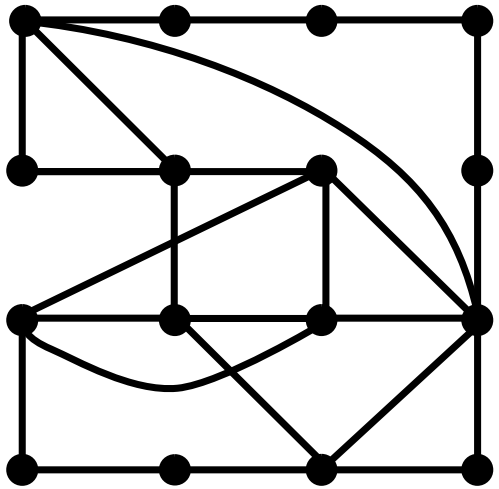


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# Example

- Is this an Eulerian graph? How do we find an Eulerian circuit?

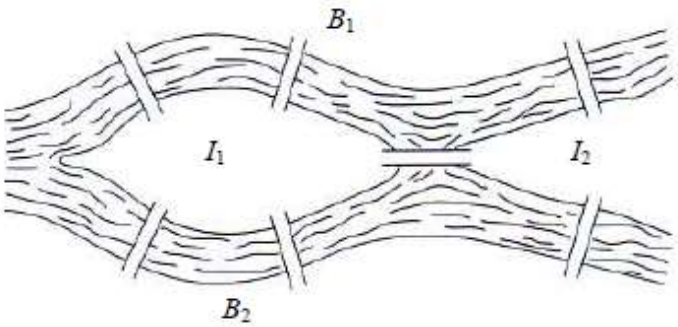




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# How did Euler become involved?

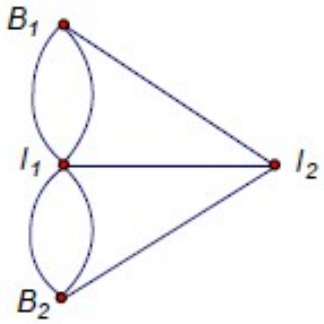
- The seven bridges of Königsberg







- Question: is there a way to walk across all the seven bridges exactly once

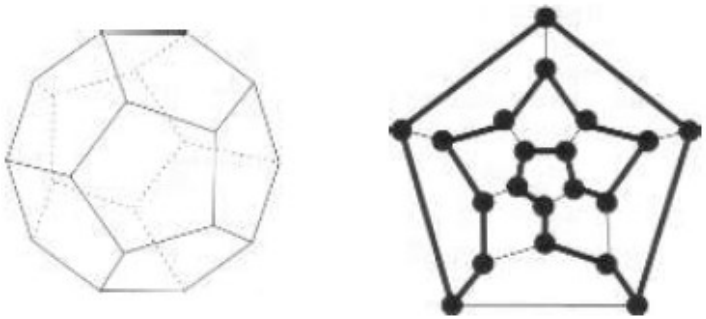





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# 2) Hamiltonian graphs

- An Eulerian graph has a trail that uses all the edges in the graph exactly once.
- What about using or visiting all the vertices exactly once?
- A **Hamiltonian cycle** in a graph $G$ is a cycle that includes every vertex exactly once. The graph $G$ is said to be **Hamiltonian** if it has a Hamiltonian cycle.



- The dodecahedron graph is Hamiltonian. A Hamiltonian cycle is shown in bold.



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# Semi-Hamiltonian graphs

- A **Hamiltonian path** is a path that includes all the vertices exactly once.
- A graph G that has a Hamiltonian path but not a Hamiltonian cycle is called a **semi–Hamiltonian graph**.
- Properties of Hamiltonian graphs
- Every Hamiltonian graph is connected.
- A tree has no cycle and hence cannot be Hamiltonian.
- A bipartite graph with an odd number of vertices cannot be Hamiltonian.
- For $n \geq 3$, the cycle graph $C_n$ and the complete graph $K_n$ are Hamiltonian.
- For $n \geq 2$, the complete bipartite graph $K_{n,n}$ is Hamiltonian.
- Since loops and multiple edges don't matter when it comes to the property of being Hamiltonian, we will consider only simple connected graphs.



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# Eulerian versus Hamiltonian

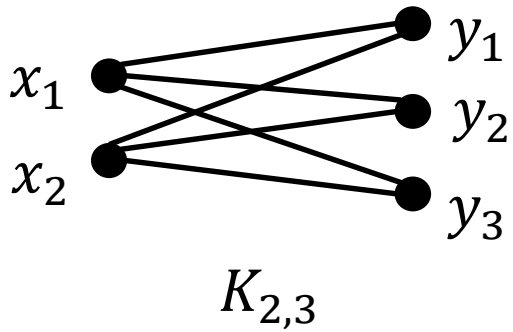
- Eulerian and semi–Eulerian graphs have easy characterisations. We just look at how many vertices of odd degrees there are in the graph.
- Is there a easy way to determine whether a graph is Hamiltonian or semi–Hamiltonian?
- Necessary condition for a graph to be Hamiltonian.
- **Theorem 2.2**
- If $G$ is a simple Hamiltonian graph, then for each $S \subseteq V(G)$, the number of components of $G \setminus S$ is at most $|S|$.
- **Proof**
- Let $C$ be a Hamiltonian cycle of $G$, viewed as a subgraph of $G$. Because $C$ is a subgraph of $G$ that includes all of the vertices of $G$, the number of components of $G \setminus S$ is at most the number of components of $C \setminus S$. Since $C$ is a cycle, the number of components of $C \setminus S$ is at most $|S|$.



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# Application of Theorem 2.2

- Consider $K_{2,3}$




$K_{2,3}$

- Is it Hamiltonian?
- What is $K_{2,3} \setminus S$ where $S = \{X_1, x_2\}$?
- $|S| = 2$ but $K_{2,3} \setminus S$ has three components (three isolated vertices). So the graph is not Hamiltonian.
- The complete bipartite graph $K_{m,n}$ is Hamiltonian if any only if $m = n$.



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# Hamiltonian graph: sufficient condition

- **Theorem 2.3 Ore's theorem**
- Let $G$ be a simple graph on $n \geq 3$ vertices. If for every pair of nonadjacent vertices $u$ and $v$ in $G$ we have $\deg(u) + \deg(v) \geq n$, then G is Hamiltonian.

- A proof relies on the following observation. Let $G$ be a simple graph on $n$ vertices. Suppose $u$ and $v$ are two nonadjacent vertices in $G$ such that $\deg(u) + \deg(v) \geq n$. We add the edge $uv$ to $G$ and call the resulting graph $H$. Then $H$ is Hamiltonian if and only if $G$ is Hamiltonian.

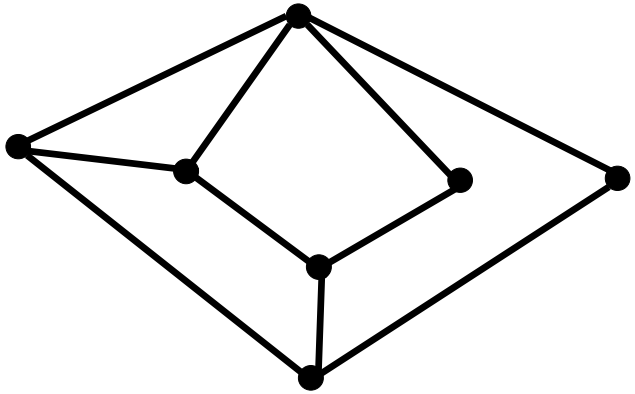
- **Corollary 2.3**
- Let $G$ be a simple graph of $n \geq 3$ vertices. If every vertex $u \in V(G)$ is such that $\deg(u) \geq {}^{n}/_{2}$, then $G$ is Hamiltonian.
- Why does corollary 2.3 follow from Theorem 2.3?



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# The usefulness of the result

- Result: suppose $u$ and $v$ are two non-adjacent vertices in $G$ such that $\deg(u) + \deg(v) \geq n$. Then $G + uv$ is Hamiltonian if and only if $G$ is Hamiltonian.
- Is the following graph Hamiltonian?





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# 3) Shortest paths

- In practical applications, it is often of interest to find the shortest route between two locations or nodes or vertices. If we use a graph to model the situation, this amounts to finding a path with the shortest length between two vertices.

- In a connected graph $G$, among all the possible paths between vertex $u$ and $v$, a path with the smallest length is called a **shortest path** from $u$ to $v$. The **length** of a path between two vertices is the number of edges on the path. The length of the shortest path is called the **distance** from $u$ to $v$. Sometimes, we write $d(u, v)$ or $\mathrm{dist}(u, v)$ for the distance between $u$ to $v$.



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# How to find a shortest path from $u$ to $v$

- How to find a shortest path from $u$ to $v$ in a connected graph?
- Idea: make a list of vertices that are closest to $u$ and growth the list iteratively starting with $u$. At step $i$, we list the vertices that are at a distance of $i$ from $u$.
- Step 1. Set $i = 0$ and $L(0) = \{u\} = M(0)$.
- Step 2. Let $L(i+1)$ be the vertices that are adjacent to some vertex in $L(i)$. Update $M(i+1) = M(i) \cup L(i+1)$.
- Step 3. Are there any vertex not $M(i+1)$? If not, **stop**.
- Otherwise set $i \longmapsto i+1$. Go to Step 2.

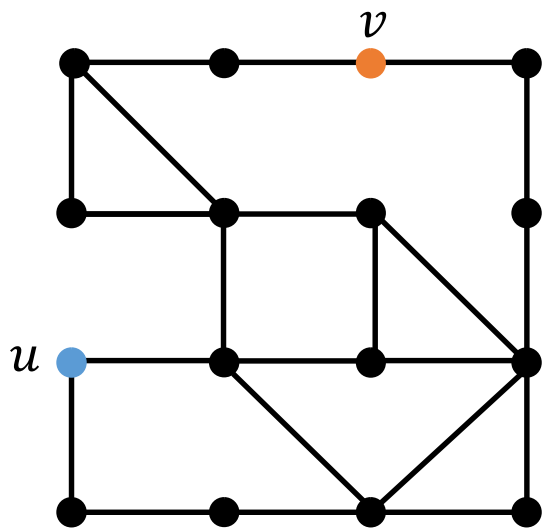
- When the process ends, we found the distances of every vertex from $u$. If vertex $v \in L(k)$ then the shortest path from $u$ to $v$ has length $k$.



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# Example

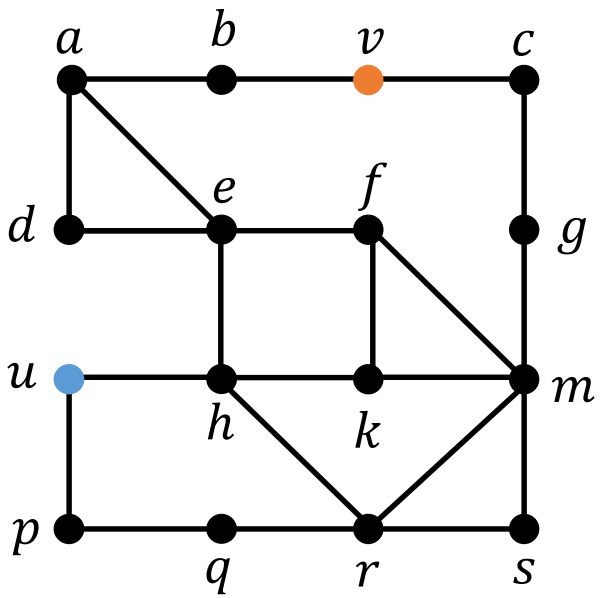
- In the following graph, we would like to find a shortest path from $u$ to $v$ and also determine the distance between $u$ and $v$.



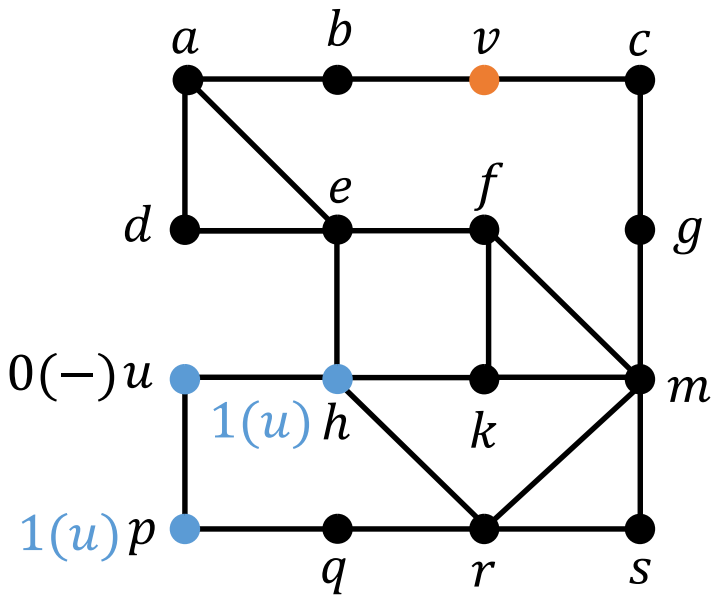



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# Example: iterations 0 and 1




$L(0) = \{u\}$
$M(0) = \{u\}$



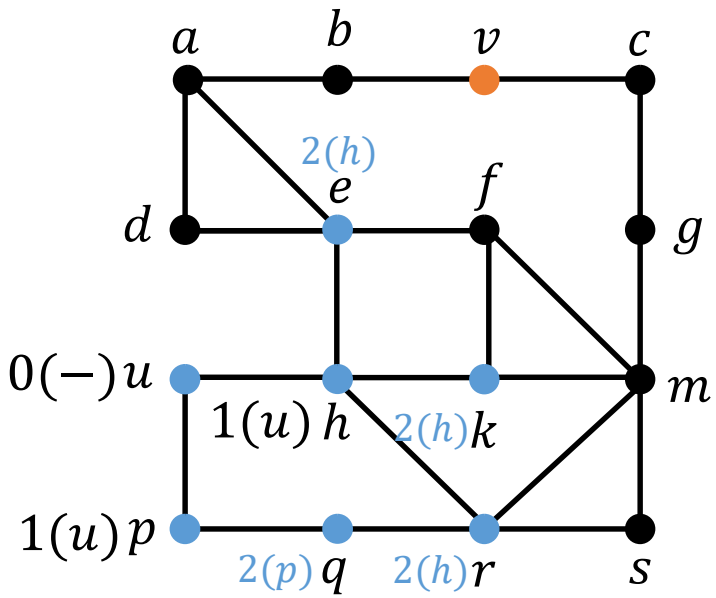

$L(1) = \{h, p\}$
$M(1) = \{u, h, p\}$



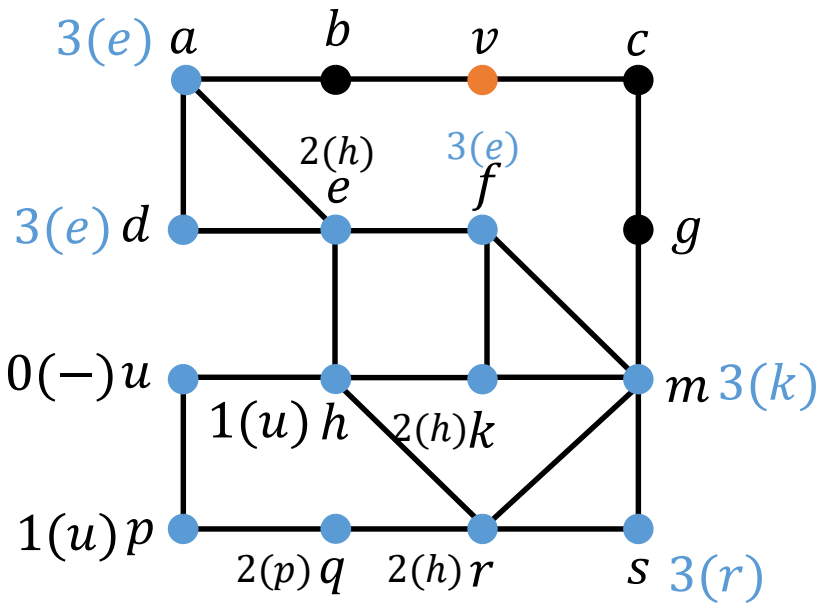
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# Example: iterations 2 and 3

When there is a choice, it may be made arbitrarily




$L(2) = \{e, k, q\}$
$M(2) = \{u, h, p, e, k, q\}$

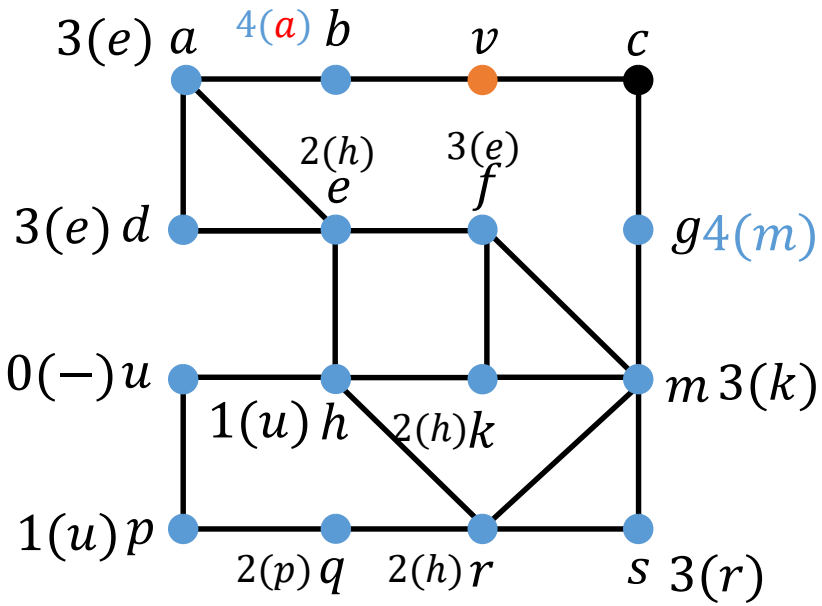



$L(3) = \{a, f, m, r, s\}$
$M(3) = \{u, h, p, e, k, q, a, f, m, r, s\}$

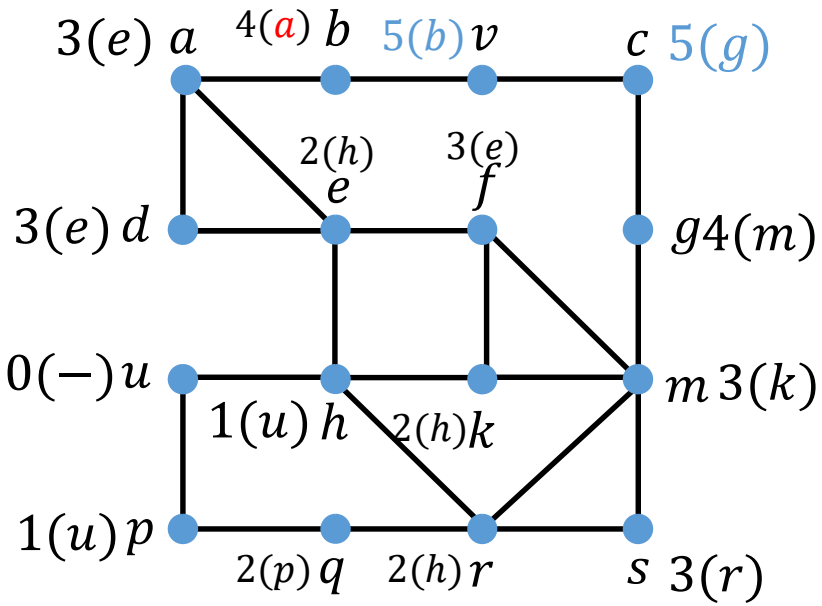


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# Example: iterations 4 and 5




$L(4) = \{b, g\}$
$M(4) = \{u, h, p, e, k, q, a, f, m, r, s, b, g\}$



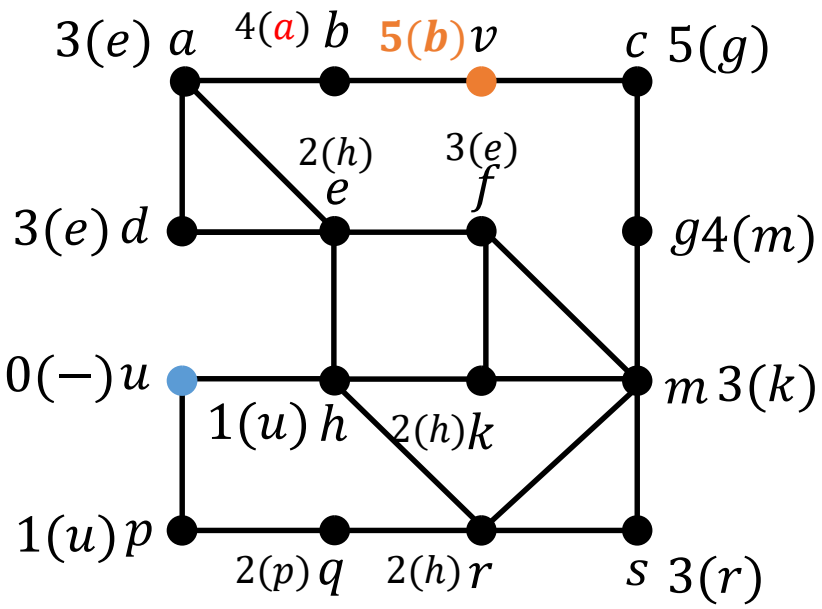

$L(5) = \{v, c\}$
$M(5) = \{u, h, p, e, k, q, a, f, m, r, s, b, g, v, c\}$



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# Example: shortest paths

- In the following graph, we would like to find a shortest path from $u$ to $v$ and also determine the distance between $u$ and $v$.




- A shortest path from $u$ to $v$: $\boldsymbol{uheabv}$
- Distance between $u$ and $v$, $d(u, v) = 5$

- The algorithm gives the shortest paths and distances from $u$.

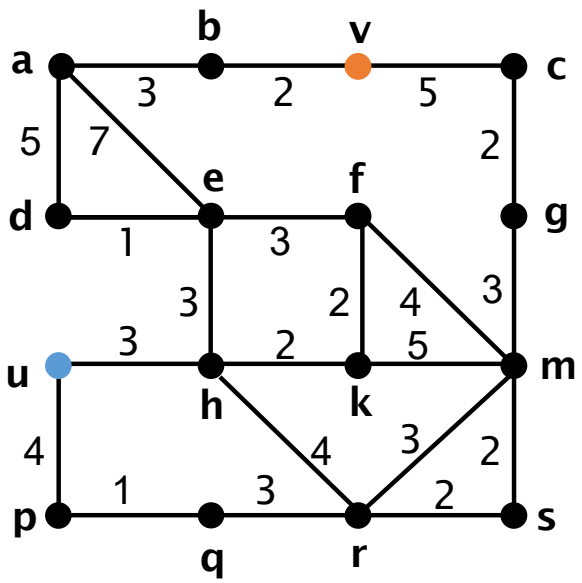
- For example:
- A shortest path from $u$ to $s$: $\boldsymbol{uhrs}$
- Distance between $u$ to $s$, $d(u, s) = 3$



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# Shortest paths in weighted graphs

- In the earlier example, every edge has a distance or weight of 1 unit. In using a graph to model a real world situation, the weights on the edges are typically not all the same.




- For a given path, say **adefkh**, its **weight** is the sum of the weights of the edges in the path. The weight of **adefkh** is 5 + 1 + 3 + 2 + 2 = 13.

- A path between two vertices with the smallest possible weight is called a **shortest path** between the two vertices. The weight of a shortest path between two vertices is called the **distance** between the two vertices.

- We wish to find a path from **u** to **v** that has the smallest weight.



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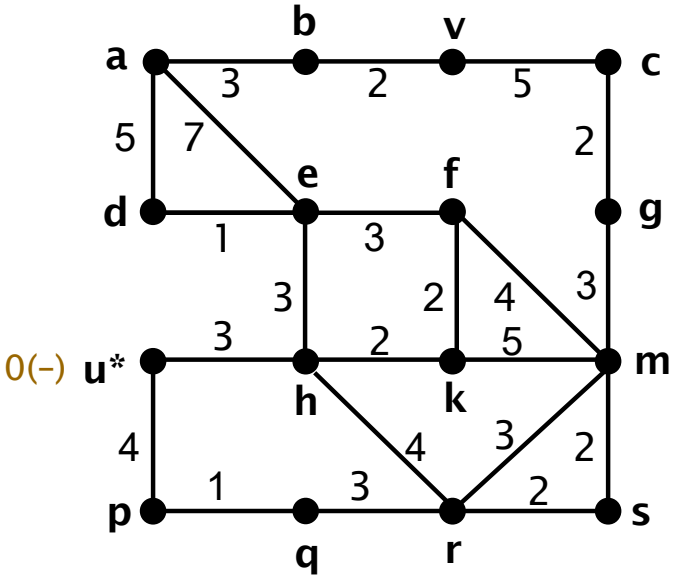
# Shortest paths in weighted graphs: Dijkstra's algorithm

- Given a weighted connected graph, we wish to find a shortest path between two vertices $u$ and $v$.
- In the recipe, we successively build the set $P$ of labelled vertices whose shortest paths from $u$ are known.
- Step 1. Assign label $0(-)$ to $u$. The vertex $u$ has 0 distance and no predecessor. Label all the other vertices $\infty(u)$. Set $P = \{u\}$
- Step 2. Find all the vertices not in $P$ that is adjacent to a vertex in $P$. Suppose $y \notin P$ is adjacent to $x_1, \cdots, x_k \in P$. Let $x_i$ have the label $d_i(t)$. Here $d_i$ is the distance of $x_i$ from $u$ and $t_i$ is the vertex preceding $x_i$ in a shortest path from $u$ to $x_i$. Let the weight of the edge $x_i y$ be $w_i$. Set $l = \min\{d_i + w_i : i = 1, \cdots, k\}$. Suppose $l = d_j + w_j =$ for index $j$. Give $y$ a temporary label $l(x_j)$. Make the temporary labels for all vertices not in $P$ that is adjacent to a vertex in $P$. Pick all the vertices with the smallest label and include them in $P$.
- Step 3. If all the vertices are in $P$, stop. Otherwise go to Step 2.

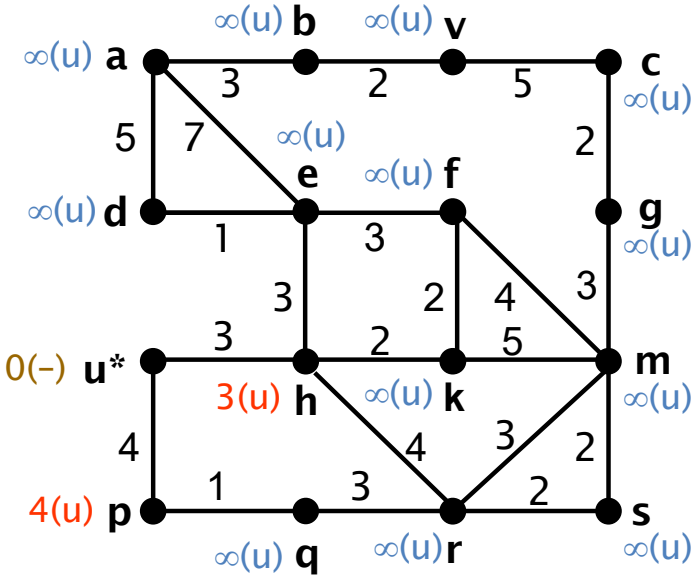


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# Example: iteration 1




1) $P = \{\mathbf{u}\}$

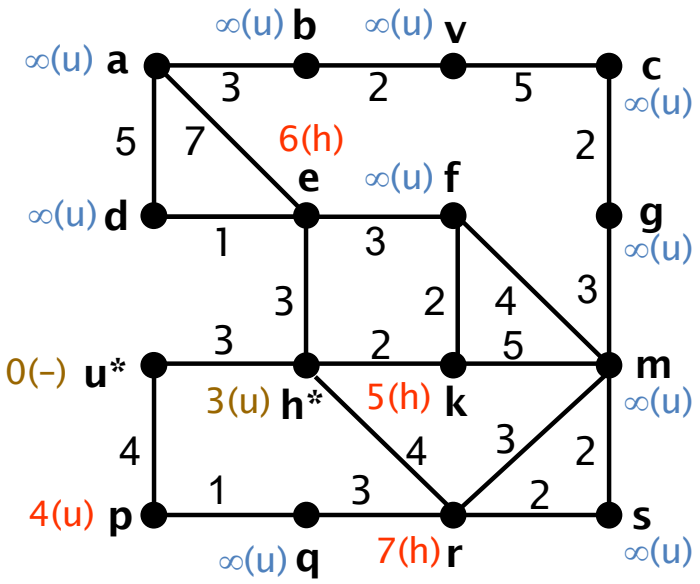



2) $P = \{\mathbf{u}, \mathbf{h}\}$



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# Example: iteration 2




3) $P$ = {**u, h, p**}

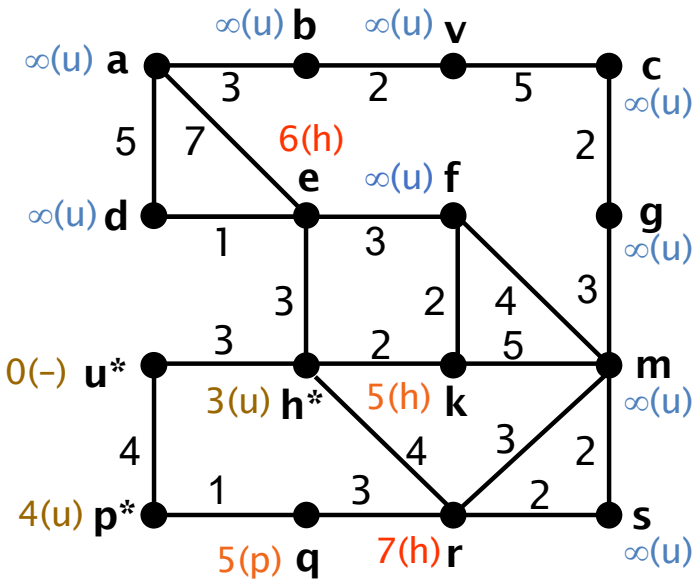
| Vertex | Old distance | New distance | Minimum (old and new) |
|---|---|---|---|
| e | ∞ | 3 + 3 | 6 |
| k | ∞ | 3 + 2 | 5 |
| r | ∞ | 3 + 4 | 7 |
| p | 4 | 4 | 4 |

- Which vertex not in $P$ has the smallest label?
- Put **p** in $P$ and repeat process.



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# Example: iteration 3




4) $P$ = {**u, h, p, k, q**}

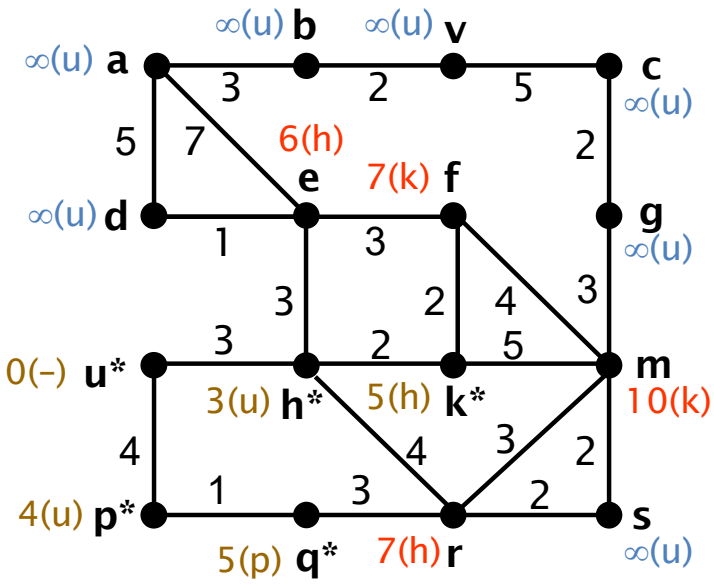
| Vertex | Old distance | New distance | Minimum (old and new) |
|---|---|---|---|
| e | 6 | 6 | 6 |
| k | 5 | 5 | 5 |
| r | 7 | 7 | 7 |
| q | ∞ | 4 + 1 | 5 |

- Both k and q have the smallest distance
- Put **k** and **q** in $P$ and repeat process.



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# Example: iteration 4




5) $P$ = {**u, h, p, k, q, e**}

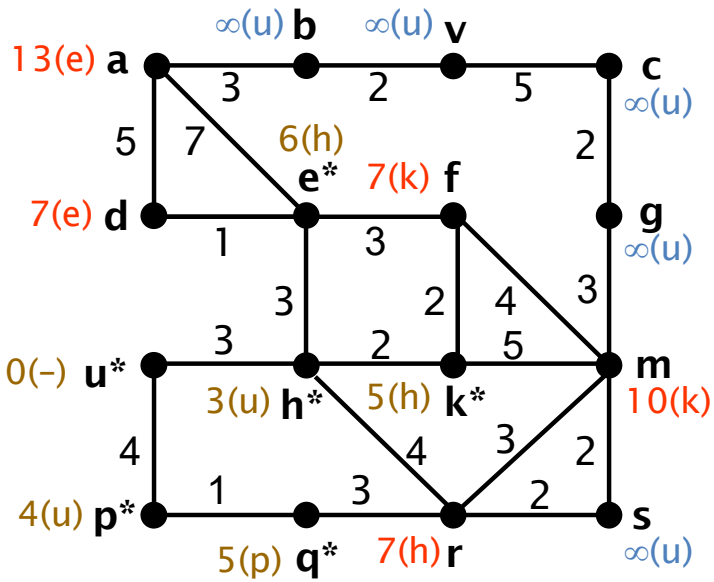
| Vertex | Old distance | New distance | Minimum (old and new) |
|---|---|---|---|
| e | 6 | 6 | 6 |
| r | 7 | 3 + 4 | 7 |
| f | ∞ | 5 + 2 | 7 |
| m | ∞ | 5 + 5 | 10 |

- Put **e** in $P$ and repeat process.



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# Example: iteration 5




6) $P$ = {**u, h, p, k, q, e, r, f, d**}

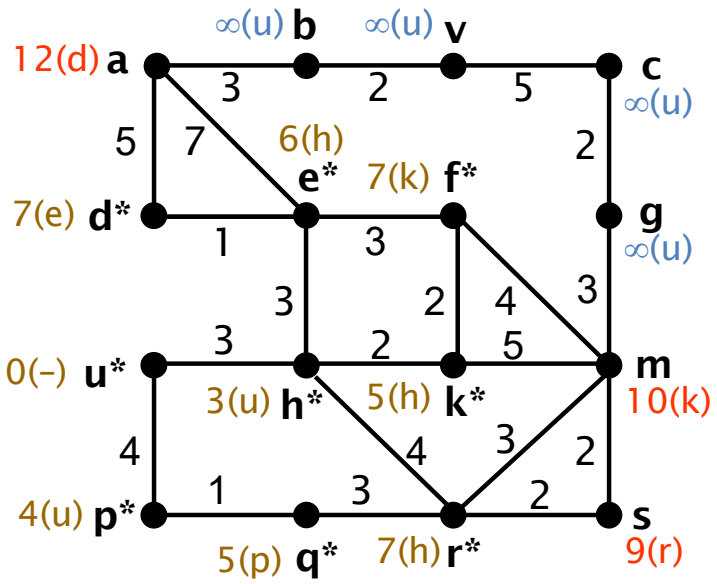
| Vertex | Old distance | New distance | Minimum (old and new) |
|---|---|---|---|
| r | 7 | 7 | 7 |
| f | 7 | 7 | 7 |
| m | 10 | 10 | 10 |
| d | ∞ | 6 + 1 | 7 |
| a | ∞ | 6 + 7 | 13 |

- Vertices r, f and d have the smallest distance
- Put **r**, **f** and **d** in $P$ and repeat process.



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# Example: iteration 6




7) $P$ = {u, h, p, k, q, e, r, f, d, s}

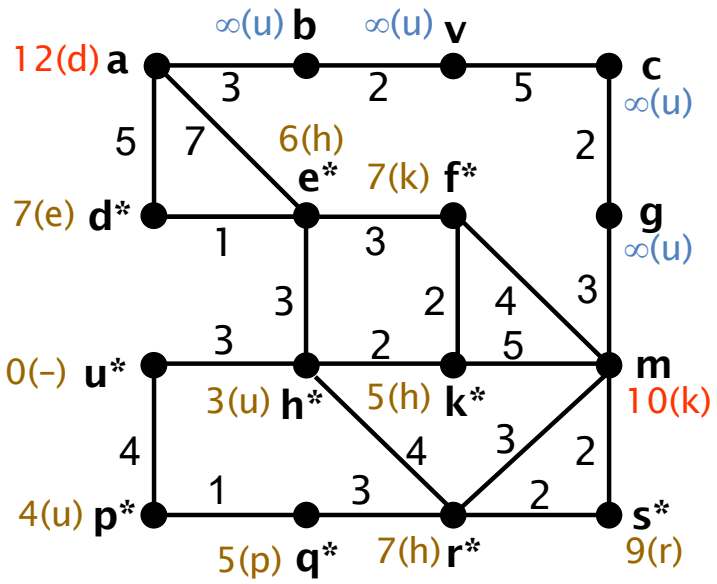
| Vertex | Old distance | New distance | Minimum (old and new) |
|---|---|---|---|
| m | 10 | 10 | 10 |
| a | 13 | 7 + 5 | 12 |
| s | ∞ | 7 + 2 | 9 |

- Put **s** in $P$ and repeat process.

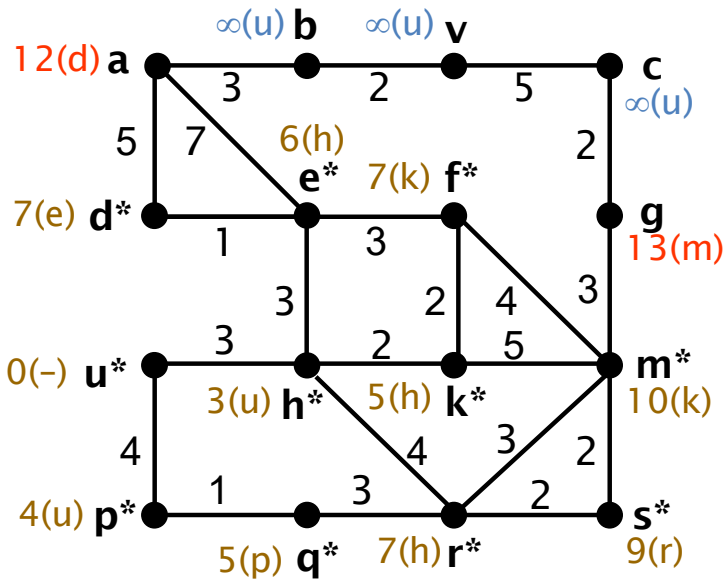


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# Example: iterations 7 and 8




8) $P$ = {u, h, p, k, q, e, r, f, d, s, m}

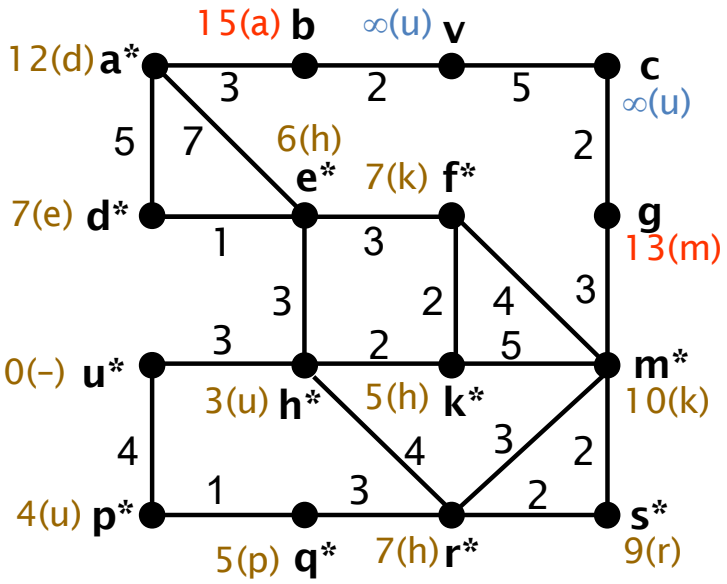



9) $P$ = {u, h, p, k, q, e, r, f, d, s, m, a}

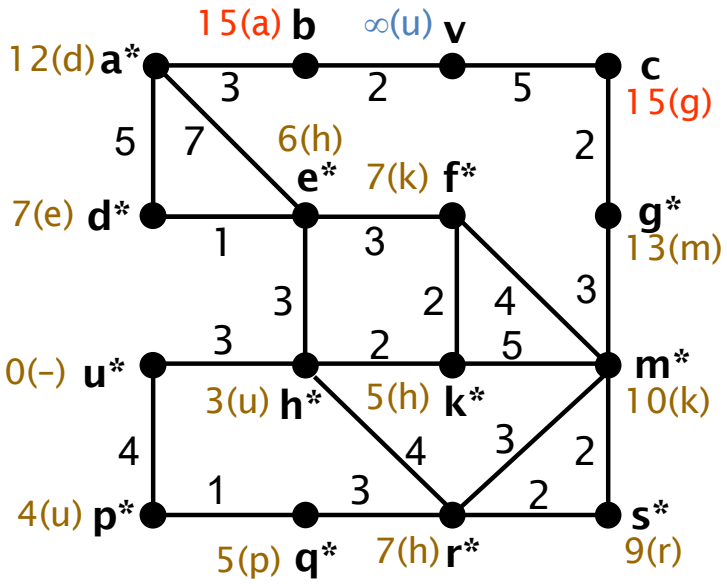


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# Example: iterations 9 and 10




10) $P$ = {u, h, p, k, q, e, r, f, d, s, m, a, g}

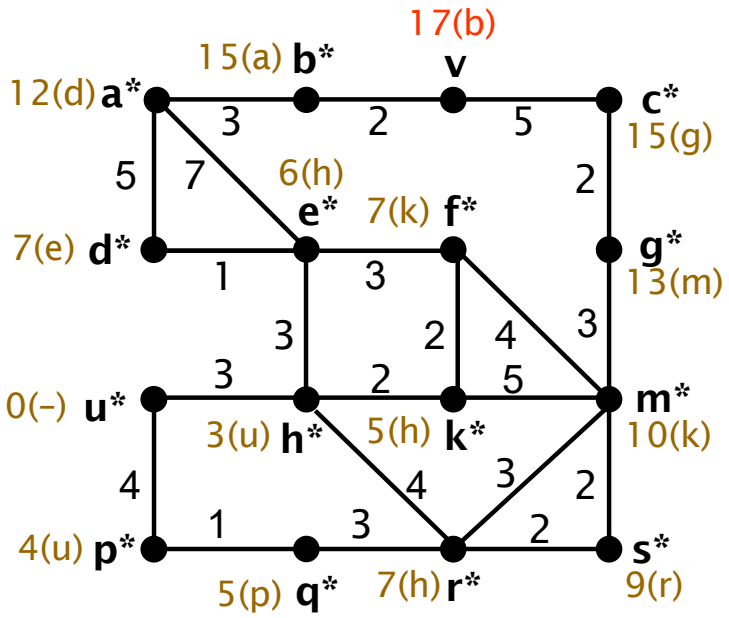



11) $P$ = {u, h, p, k, q, e, r, f, d, s, m, a, g, b, c}



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# Example: iteration 11




- The algorithm ends when all the vertices are in $P$
- A shortest path from u to v is uhedabv and its weight or length is 3 + 3 + 1 + 5 + 3 + 2 = 17.

- The recipe produces the shortest paths from u to the other vertices. For example, a shortest path from u to s is uhrs with a weight of 3 + 4 + 2 = 9.

12) $P$ = {**u, h, p, k, q, e, r, f, d, s, m, a, g, b, c, v**}



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# Summary of the stages

| Labels and Predecessors | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Stage | u | a | b | c | d | e | f | g | h | k | m | p | q | r | s | v | Vertex added |
| 1 | 0(-) | ∞(u) | ∞(u) | ∞(u) | ∞(u) | ∞(u) | ∞(u) | ∞(u) | ∞(u) | ∞(u) | ∞(u) | ∞(u) | ∞(u) | ∞(u) | ∞(u) | ∞(u) | u |
| 2 | 0(-) | ∞(u) | ∞(u) | ∞(u) | ∞(u) | ∞(u) | ∞(u) | ∞(u) | 3(u) | ∞(u) | ∞(u) | 4(u) | ∞(u) | ∞(u) | ∞(u) | ∞(u) | h |
| 3 | 0(-) | ∞(u) | ∞(u) | ∞(u) | ∞(u) | 6(h) | ∞(u) | ∞(u) | 3(u) | 5(h) | ∞(u) | 4(u) | ∞(u) | 7(h) | ∞(u) | ∞(u) | p |
| 4 | 0(-) | ∞(u) | ∞(u) | ∞(u) | ∞(u) | 6(h) | ∞(u) | ∞(u) | 3(u) | 5(h) | ∞(u) | 4(u) | 5(p) | 7(h) | ∞(u) | ∞(u) | k, q |
| 5 | 0(-) | ∞(u) | ∞(u) | ∞(u) | ∞(u) | 6(h) | 7(k) | ∞(u) | 3(u) | 5(h) | 10(k) | 4(u) | 5(p) | 7(h) | ∞(u) | ∞(u) | e |
| 6 | 0(-) | 13(e) | ∞(u) | ∞(u) | 7(e) | 6(h) | 7(k) | ∞(u) | 3(u) | 5(h) | 10(k) | 4(u) | 5(p) | 7(h) | ∞(u) | ∞(u) | d, f, r |
| 7 | 0(-) | 12(e) | ∞(u) | ∞(u) | 7(e) | 6(h) | 7(k) | ∞(u) | 3(u) | 5(h) | 10(k) | 4(u) | 5(p) | 7(h) | 9(r) | ∞(u) | s |
| 8 | 0(-) | 12(e) | ∞(u) | ∞(u) | 7(e) | 6(h) | 7(k) | ∞(u) | 3(u) | 5(h) | 10(k) | 4(u) | 5(p) | 7(h) | 9(r) | ∞(u) | m |
| 9 | 0(-) | 12(e) | ∞(u) | ∞(u) | 7(e) | 6(h) | 7(k) | 13(m) | 3(u) | 5(h) | 10(k) | 4(u) | 5(p) | 7(h) | 9(r) | ∞(u) | a |
| 10 | 0(-) | 12(e) | 15(a) | ∞(u) | 7(e) | 6(h) | 7(k) | 13(m) | 3(u) | 5(h) | 10(k) | 4(u) | 5(p) | 7(h) | 9(r) | ∞(u) | g |
| 11 | 0(-) | 12(e) | 15(a) | 15(g) | 7(e) | 6(h) | 7(k) | 13(m) | 3(u) | 5(h) | 10(k) | 4(u) | 5(p) | 7(h) | 9(r) | ∞(u) | b, c |
| 12 | 0(-) | 12(e) | 15(a) | 15(g) | 7(e) | 6(h) | 7(k) | 13(m) | 3(u) | 5(h) | 10(k) | 4(u) | 5(p) | 7(h) | 9(r) | 17(b) | v |



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# Section 3: Directed graphs or digraphs

- 1) Digraphs or directed graphs
- 2) In and out degrees of a vertex
- 3) Isomorphic digraphs
- 4) Paths and cycles in digraphs
- 5) Eulerian and Hamiltonian digraphs



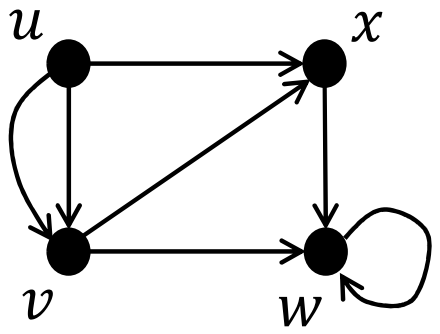
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# 1) Directed graphs or digraphs

- Sometimes an orientation or direction has to be placed on the edges to model a given problem. For instance, there are roads that one-way or electric circuits where current flows in a particular direction.
- A **directed graph** or **digraph** $D$ consists of a set of vertices and a set of arcs, where each arc joins a pair of vertices in a specific direction. The set of vertices of $D$ is denoted by $V(D)$ or simply $V$ if there is no confusion, and the set of arcs is denoted by $E(D)$ or $E$. The digraph $D$ may be written as $D(V, E)$.
- An arc $e$ joining two vertices $u$ and $v$ with direction from $u$ to $v$ can be denoted by $e = uv$ or $(u, v)$. The direction matters and so $uv \neq vu$ and $(u, v) \neq (v, u)$.

- Vertex set, $V(D) = \{u, v, w, x\}$
- Edge set, $E(D) = \{uv, uv, ux, vw, vx, ww, xw\}$




Digraph $D$

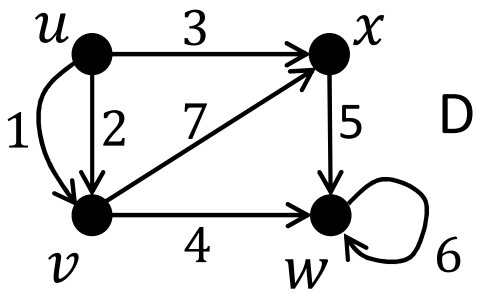


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# Loops, multiple arcs, simple digraphs

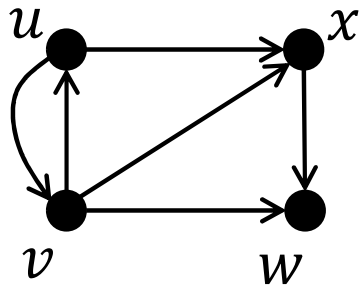
- In a digraph, a **loop** is an arc joining a vertex to itself. **Multiple arcs** occur when two or more arcs joining a pair of vertices in the same direction.
- A digraph with no loops or multiple arcs is a **simple digraph**.

- In digraph $D$, arcs $1$ and $2$ are multiple arcs. Arc $6$ is a loop.




Digraph $D$

- Is digraph $H$ a simple digraph?



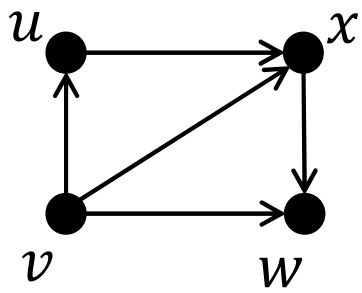

Digraph $H$



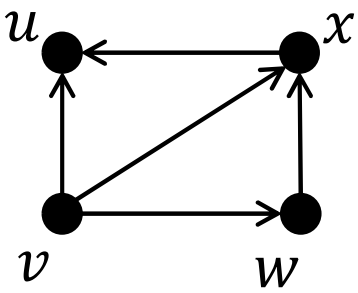
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# Oriented graphs

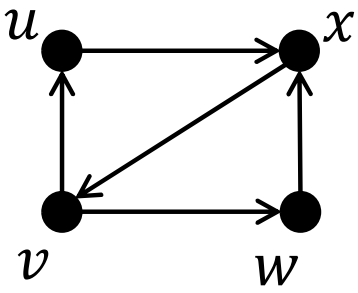
- In a digraph, if for each pair of distinct vertices $u$ and $v$, at most one of $uv$ and $vu$ is an arc, then the resulting digraph is an **oriented graph**.
- Thus an oriented graph $D$ is obtained by assigning a direction to each edge of some simple graph $G$. The digraph $D$ is also called an **orientation** of $G$.



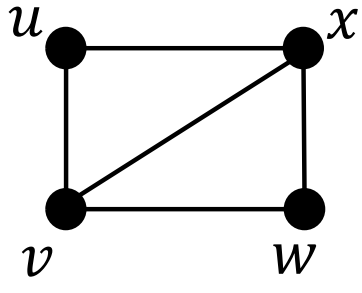

Digraph $M1$




Digraph $M2$




Digraph $M3$




Graph $G$

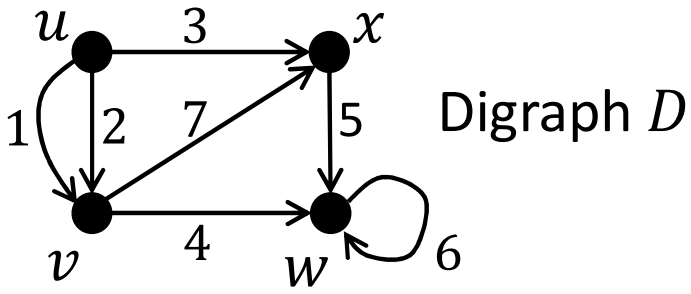
- Digraphs $M1$, $M2$ and $M3$ are oriented graphs. They are orientations of the simple graph $G$.
- Is digraph $H$ on the previous slide an oriented graph?



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# Adjacent to/from, incident to/from

- If $e = uv$ is an arc in a digraph $D$, then $u$ is said to be **adjacent to** $v$ and $v$ is **adjacent from** $u$. The arc $e$ is said to be **incident from** $u$ and **incident to** $v$. We also say that $u$ is **incident to** $e$, and $v$ is **incident from** $e$.




Digraph $D$

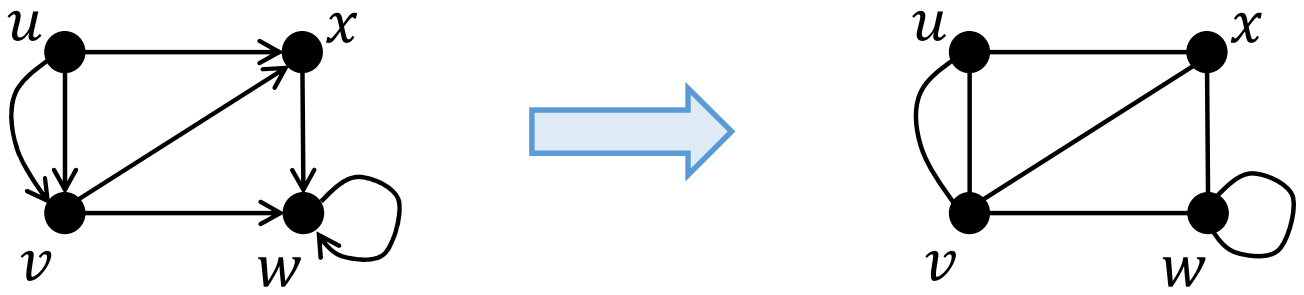
- In digraph $D$, we say that vertex $u$ is adjacent to $v$.
- Vertex $x$ is adjacent from $v$.
- Vertex $u$ is incident to arcs 1, 2 and 3.
- Vertex $w$ is incident from arcs 4, 5 and 6.
- Arc 7 is incident from $v$ and incident to $x$.
- Arc 6 is incident to, as well as incident from, $w$.



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# Underlying graph to a digraph

- The **underlying graph** of a digraph $D$ is the graph obtained by replacing any arc by the corresponding undirected edge.
- To obtain the underlying graph, we simply remove the arrows from the arcs.
- For example,




Digraph $D$ Underlying graph

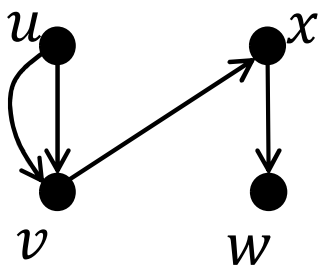


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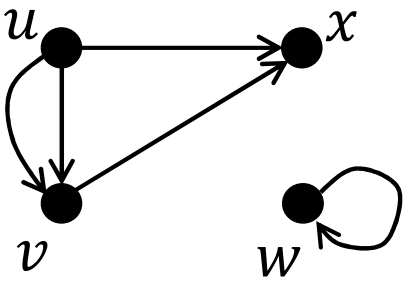
# Sub-digraphs

- A digraph $C\big(V(C), E(C)\big)$ is called a **sub-digraph** of digraph $D\big(V(D),\ E(D)\big)$ if $V(C)\ \subseteq V(D)$ and $E(C)\ \subseteq E(D)$.
- Furthermore if $V(C) = V(D)$, the sub-digraph $C$is called a **spanning sub-digraph** of $D$.

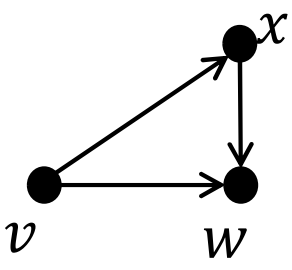
- The digraphs $C1$, $C2$ and $C3$ below are sub-digraphs of $D$.



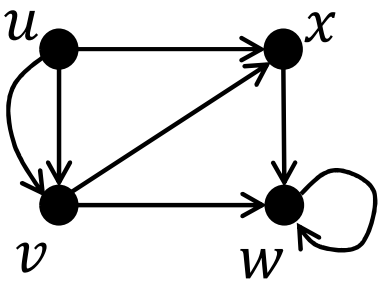
Digraph $C1$



Digraph $C2$



Digraph $C3$



Digraph $D$

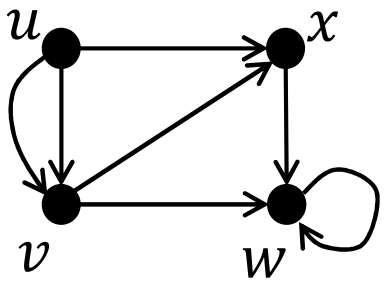
- Which sub-digraphs are spanning sub-digraphs?



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# 2) In and out degrees of a vertex

- In a digraph $D$, the out-degree of a vertex $v$ is the number of arcs incident from $v$, and is denoted by $\text{outdeg}(v)$. The in-degree of $v$ is the number of arcs incident to $v$, and is denoted by $\text{indeg}(v)$.
- For the digraph $D$, the in-degrees and out-degrees of the vertices are as follows:




Digraph $D$

- Vertex $u$: $\text{indeg}(u) = 0$ and $\text{outdeg}(u) = 3$.
- Vertex $v$: $\text{indeg}(v) = 2$ and $\text{outdeg}(v) = 2$.
- Vertex $w$: $\text{indeg}(w) = 3$ and $\text{outdeg}(w) = 1$.
- Vertex $x$: $\text{indeg}(x) = 2$ and $\text{outdeg}(x) = 1$.
- What do we notice?



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# Handshake lemma for directed graphs

- **Theorem 3.1 Handshaking Lemma**
- In a digraph $D$ with $m$ arcs, the sum of all the vertex out-degrees and the sum of all the vertex in-degrees are both equal to $m$, the number of the arcs. That is,

$$\sum_{u \in V(D)} \text{outdeg}(u) = \sum_{u \in V(D)} \text{indeg}(u) = m$$

- **Proof**
- Each arc, say $e = uv$ contributes exactly 1 to the out-degree of $u$ and 1 to the in-degree of $v$. Hence the result.

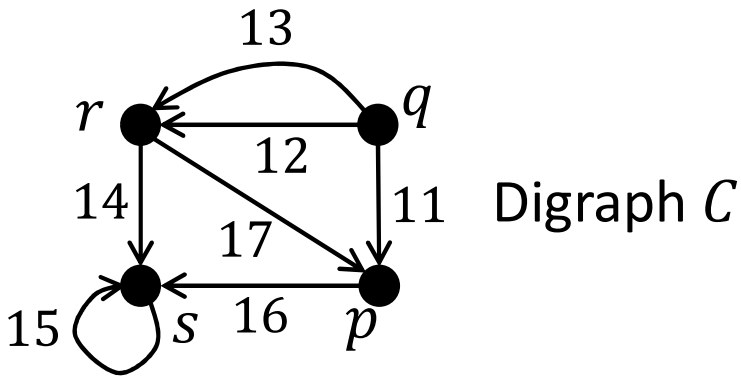


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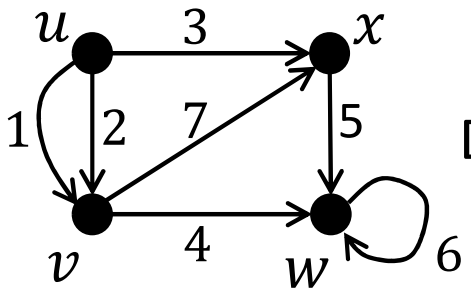
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# 3) Isomorphic digraphs

- Two digraphs $C$ and $D$ are **isomorphic** if there exists a one-to-one correspondence (bijection) $\varphi$ from $V(C)$ to $V(D)$ such $uv \in E(C)$ if and only if $\varphi(u)\varphi(v) \in E(D)$. Such a bijection is known as an **isomorphism** from $C$ to $D$.
- Are the digraphs $C$ and $D$ below isomorphic?




Digraph $C$




Digraph $D$

- Consider the map $\varphi: V(C) \longrightarrow V(D)$ where $\varphi(p) = x, \varphi(q) = u, \varphi(r) = v, \varphi(s) = w$.
- It is straightforward to verify that $\varphi$ is an isomorphism. The correspondence of the arcs are

$$11 \leftrightarrow 3,\ 12 \leftrightarrow 2,\ 13 \leftrightarrow 1,\ 14 \leftrightarrow 4,\ 15 \leftrightarrow 6,\ 16 \leftrightarrow 5,\ 17 \leftrightarrow 7$$

- Thus $C$ and $D$ are isomorphic digraphs.



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# An easy consequence

- **Theorem 3.2**

- Let φ be an isomorphism from digraphs $C$ to $D$. For an vertex $u \in V(C)$, the out-degree of $u$ and the out-degree of $\varphi(u)$ are equal. Similarly the in-degree of $u$ and the in-degree of $\varphi(u)$ are equal.
- **Proof**
- Let the out-degree of $u$ be $k$ and let the arcs $uw_1, uw_2, \cdots, uw_k$ be incident from $u$. Since $\varphi$ is an isomorphism, the arcs incident from $\varphi(u)$ in digraph $D$ are exactly $\varphi(u)\varphi(w_1), \varphi(u)\varphi(w_2), \cdots, \varphi(u)\varphi(w_k)$. Thus the out-degree of $\varphi(u)$ is $k$.
- A similar argument establishes the equality of the in-degree of $u$ and the in-degree of $\varphi(u)$. ■



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# 4) Paths and cycles in digraphs

- The notions of walk, trail and path in an undirected graph can be applied to a digraph. An obvious requirement is that the walk or trail or path must respect the direction of the associated arc.

- A **walk of length $k$** in a digraph is a succession of $k$ arcs of the form $u_1u_2, u_2u_3, \cdots, u_ku_{k=1}$. We denote this walk by $u_1u_2 \cdots u_{k+1}$ and refer to it as a **walk from $u_1$ to $u_{k+1}$**. A walk between $u$ and $v$ is sometimes denoted by **$u - v$ walk**.

- A **trail** in a digraph is a walk in which all the arcs (but not necessarily all the vertices) are distinct.

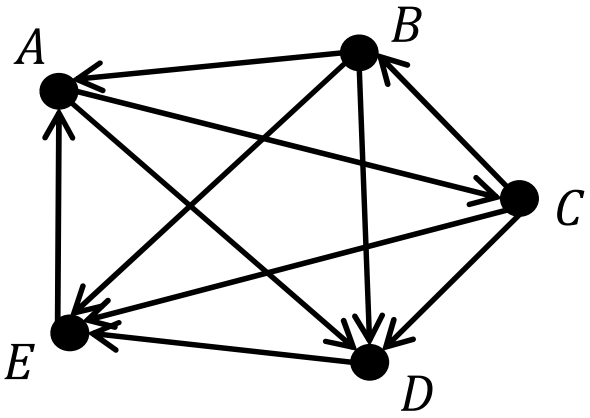
- A **path** in a digraph is a trail in which all the vertices are distinct.



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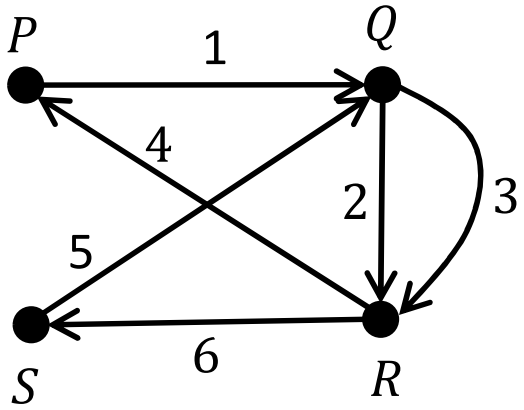
# Examples

- Consider the digraph




- $ACEADEAD$ is a walk of length 7. It is not a trail.
- $ACEAD$ is a trail but not a path.
- $CBADE$ is a path of length 4.

- Consider the digraph




- Is $RSQRPQR$ a walk?
- Is $RSQRPQR$ a trail?
- Is $R-6-S-5-Q-2-R-4-Q-3-R$ a trail?



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# More terms

- **Theorem 3.3**
- Every $u - v$ walk in a digraph contains a $u - v$ path. (why?)

- A **closed walk** in a digraph is a succession of arcs of the form $u_1u_2, u_2u_3, \cdots, u_ku_1$. A closed walk is a walk that ends up at the starting vertex.

- A **closed trail** is a closed walk in which all the arcs are different. A closed trail is also called **circuit**.

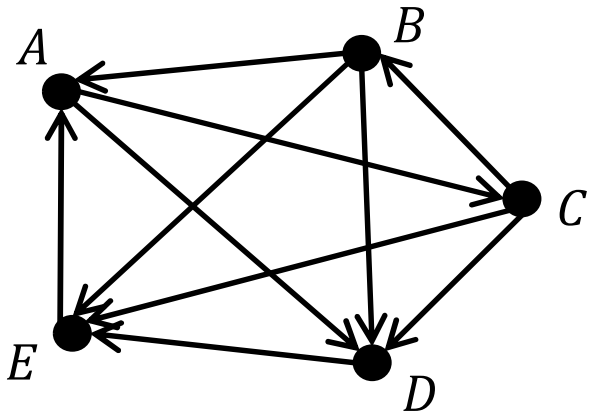
- A **cycle** is a closed trail in which all the intermediate vertices are distinct.



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# Example

- Consider the graph




- Is $BACBEACB$ a closed walk? Is it a closed trail?
- Is $ACBADEA$ a closed trail? Is it a cycle?
- The closed walk $ACBDEA$ is a circuit as well as a cycle.



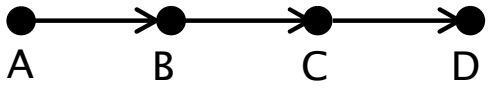
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# Connected and strongly connected digraphs

- A digraph is **connected** if its underlying graph is a connected graph, and is disconnected otherwise.

- A digraph is **strongly connected** if for each pair of vertices $u$ and $v$, there is a (directed) path from $u$ to $v$ and also a directed path from $v$ to $u$.

- Is the digraph shown below connected? Why?
- Is the digraph strongly connected? Why?




- What is the smallest number of arcs that we need to add to the digraph above to make it strongly connected?



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# When is a directed graph strongly connected?

- The following result characterizes strongly connected digraphs.

- **Theorem 3.4**
- A digraph $D$ is strongly connected if and only if $D$ contains a closed spanning walk.

- Remark: a spanning walk is one that includes all the vertices.

- **Proof**

- ($\Leftarrow$) Suppose $D$ has a closed spanning walk $W = w_1 w_2 \cdots w_n w_1$. Take any two vertices $u$ and $v$ and suppose $u = w_i$ and $v = w_j$. If $i < j$, then $w_i w_{i+1} \cdots w_j$ is a $u - v$ walk. If $i > j$, then $w_i w_{i+1} \cdots w_n w_1 \cdots w_j$ is a $u - v$ walk. Thus we can always find a $u - v$ walk. By Theorem 3.3 there is a $u - v$ path inside the $u - v$ walk. Thus $D$ is strongly connected.



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# When is a directed graph strongly connected? cont'd

- **Theorem 3.4**
- A digraph $D$ is strongly connected if and only if $D$ contains a closed spanning walk.

- ($\Rightarrow$) Suppose $D$ is strongly connected. Take any two vertices $u$ and $v$. Since there is a path from $u$ to $v$ and another path from $v$ to $u$, the concatenation of these two paths gives a closed walk starting from $u$. Take a closed walk $W$ starting from $u$ that contains the maximum number of distinct vertices. We claim that $W$ is a spanning walk.
- Suppose on the contrary that it is not. Then there is a vertex $x$ that is not in $W$. Since $D$ is strongly connected, there is a $u - x$ path, say $P$, and also a $x - u$ path, say $Q$. Consider the new walk $W'$ formed by concatenating $W, P, Q$. Now $W'$ is a closed walk starting from $u$ and has more distinct vertices than $W$. This is a contradiction. Hence $W$ is a closed spanning walk. ■



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# 5) Eulerian and Hamiltonian digraphs

- A connected digraph is **Eulerian** if it contains a circuit that includes every arc. Such a circuit is an **Eulerian circuit**.

- A connected digraph is **Hamiltonian** if it contains a cycle that includes every vertex; such a cycle is a **Hamiltonian cycle**.

- **Theorem 3.5**
- A connected digraph $D$ is Eulerian if and only if for each vertex in $D$, the out-degree equals the in-degree.

- **Corollary 3.1**
- A connected digraph $D$ is Eulerian if and only if $D$ is a disjoint union of arc–disjoint cycles in D.

- The proofs of the theorem and corollary are very similar to those for undirected graphs.

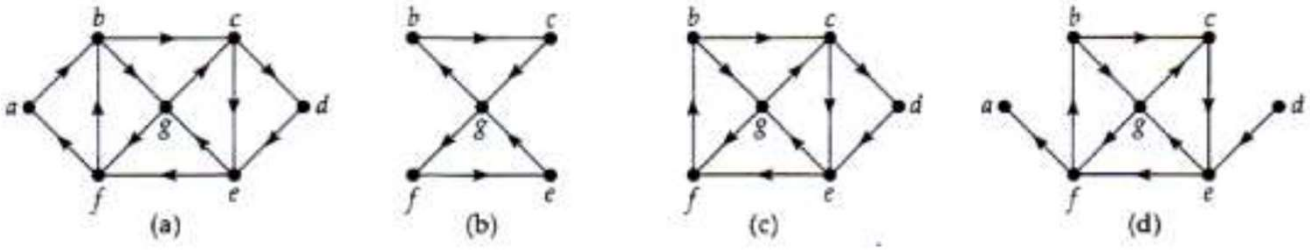


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# Examples

- Consider the four digraphs below.




- Digraph (a) is both Eulerian and Hamiltonian. An Eulerian circuit is abcdefbgcegfa and a Hamiltonian cycle is abcdegfa;
- Digraph (b) is Eulerian. An Eulerian circuit is bcgfegb; it is not Hamiltonian;
- Digraph (c) is Hamiltonian. A Hamiltonian cycle is bcdegfb; it is not Eulerian.
- Digraph (d) is neither Eulerian nor Hamiltonian.



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# Section 4: Matrix representations

- 1) Adjacency matrix
- 2) Incidence matrix



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# 1) Adjacency matrix

- We can represent a graph by drawing the vertices and edges.
- However if we want a computer to process a graph, a suitable way to capture the information is by presenting it in the form of a matrix.
- Two main types of matrices
- **Adjacency matrix**: relationship between vertices. How many edges join the two vertices?
- **Incidence matrix**: for a given vertex, which edges are incident to or from it.

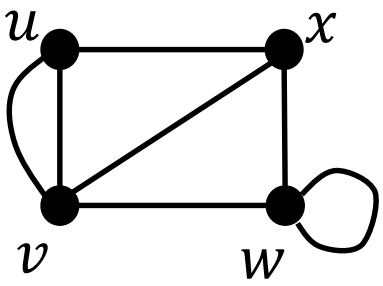


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# Adjacency matrix: undirected graph

- Let $G$ be a graph with $n$ vertices labelled by $V(G) = \{v_1, v_2, \cdots, v_n\}$. The **adjacency matrix** $A(G) = [a_{ij}]$ of $G$ is a $n \times n$ matrix where $a_{ij}$ is the number of edges joining vertices $v_i$ and $v_j$.

- For the graph below, the adjacency matrix is given on the right.




$$\begin{array}{c} \\ u \\ v \\ w \\ x \end{array} \begin{array}{c} \begin{array}{cccc} u & v & w & x \end{array} \\ \begin{bmatrix} 0 & 2 & 0 & 1 \\ 2 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \end{array}$$

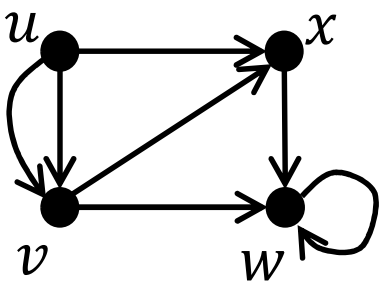
- Vertex $u$ is connected to $v$ by 2 edges and hence the $u - v$ entry is 2.
- Vertex $w$ is connected to itself by 1 edge, so $w - w$ is 1, and so forth.
- Notice: for an undirected graph, its adjacency matrix is symmetric.



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# Adjacency matrix: directed graph

- Let $D$ be a digraph with $n$ vertices labelled by $V(D) = \{v_1, v_2, \cdots, v_n\}$. The **adjacency matrix** $A(G) = [a_{ij}]$ of $D$ is a $n \times n$ matrix where $a_{ij}$ is the number of arcs from vertex $v_i$ to $v_j$.
- For the digraph below, the adjacency matrix is given on the right




$$\begin{array}{c} \\ u \\ v \\ w \\ x \end{array} \begin{array}{c} \begin{array}{cccc} u & v & w & x \end{array} \\ \begin{bmatrix} 0 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

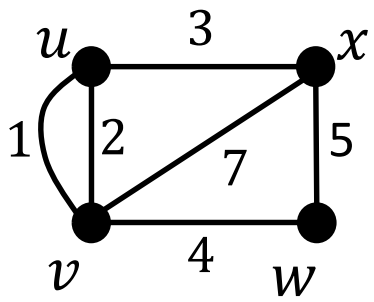
- The $u - v$ entry is 2 showing that there are two arcs from $u$ to $v$.
- The $v - u$ entry is 0 showing there is no arc from $v$ to $u$.
- There is an entry of 1 for the $w - w$ position as there is a loop at $w$.
- Is the adjacency matrix symmetric?



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# 2) Incidence matrix: undirected graph

- For the incidence matrix, we consider graphs without loops.
- Let $G$ be a graph without loops, with vertex set $V(G) = \{v_1, v_2, \cdots, v_n\}$ and edge set by $E(G) = \{e_1, e_2, \cdots, e_m\}$. The incidence matrix $B(G) = [b_{ij}]$ of $G$ is a $n \times m$ matrix where $b_{ij}$ is 1 if vertex $v_i$ is incident to edge $e_j$ and 0 otherwise.
- The incidence matrix for the graph below is given on the right.




$$\begin{array}{c} \\ u \\ v \\ w \\ x \end{array} \begin{array}{c} \begin{array}{ccccccc} 1 & 2 & 3 & 4 & 5 & 7 \end{array} \\ \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix} \end{array}$$

- What properties does the incidence matrix have?

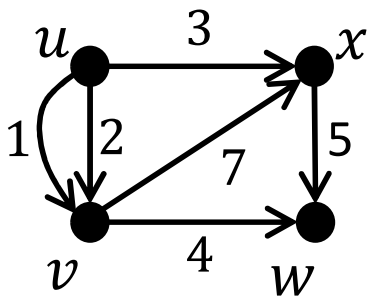


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# Incidence matrix: directed graph

- Let $D$ be a digraph without loops, with vertex set $V(D) = \{v_1, v_2, \cdots, v_n\}$ and arc set $E(D) = \{e_1, e_2, \cdots, e_m\}$. The **incidence matrix** $B(D) = [b_{ij}]$ of $D$ is a $n \times m$ matrix where $b_{ij}$ is 1 if arc $e_j$ is incident from vertex $v_i$, $-1$ if arc $e_j$ is incident to vertex $v_i$ and 0 otherwise.

- The incidence matrix for the digraph below is given on the right.




$$\begin{array}{c} \\ u \\ v \\ w \\ x \end{array} \begin{array}{c} \begin{array}{ccccccc} 1 & & 2 & 3 & 4 & 5 & 7 \end{array} \\ \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ -1 & -1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & -1 & -1 & 0 \\ 0 & 0 & 1 & 0 & 1 & -1 \end{bmatrix} \end{array}$$

- What properties does the incidence matrix have?



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# What we covered today

- **Key concepts and definitions**
- Graphs, vertices, edges, subgraphs, degree of a vertex, isomorphic graphs, paths, cycles, connected graphs, components, regular graphs, bipartite graphs, trees, complement of a simple graph
- Walks, trails, circuits, paths, Eulerian graphs, Hamiltonian cycles,
- **Results**
- When is a graph/digraph Eulerian?
- Sufficient conditions for Hamiltonian.
- **Recipes**
- Finding the shortest path in a weighted graph.
- **Questions**?



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